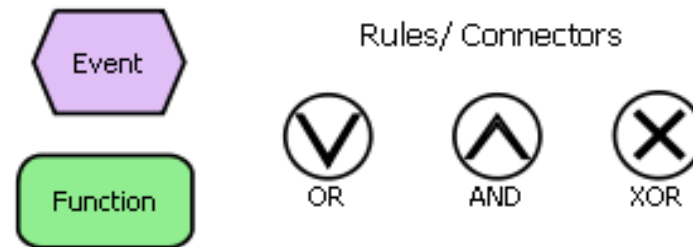


EXKURZE DO SPECIFIK MODELOVACÍCH JAZYKŮ

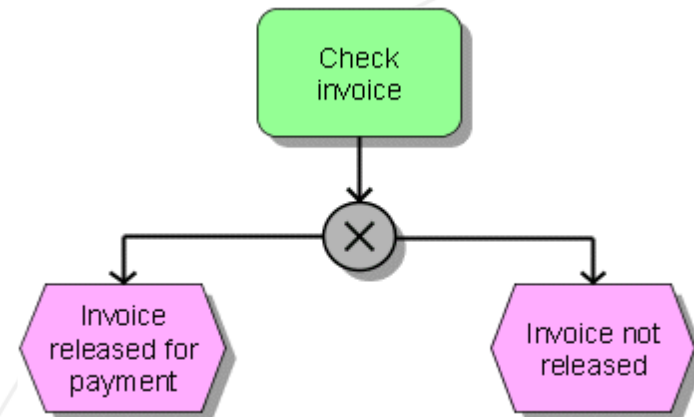
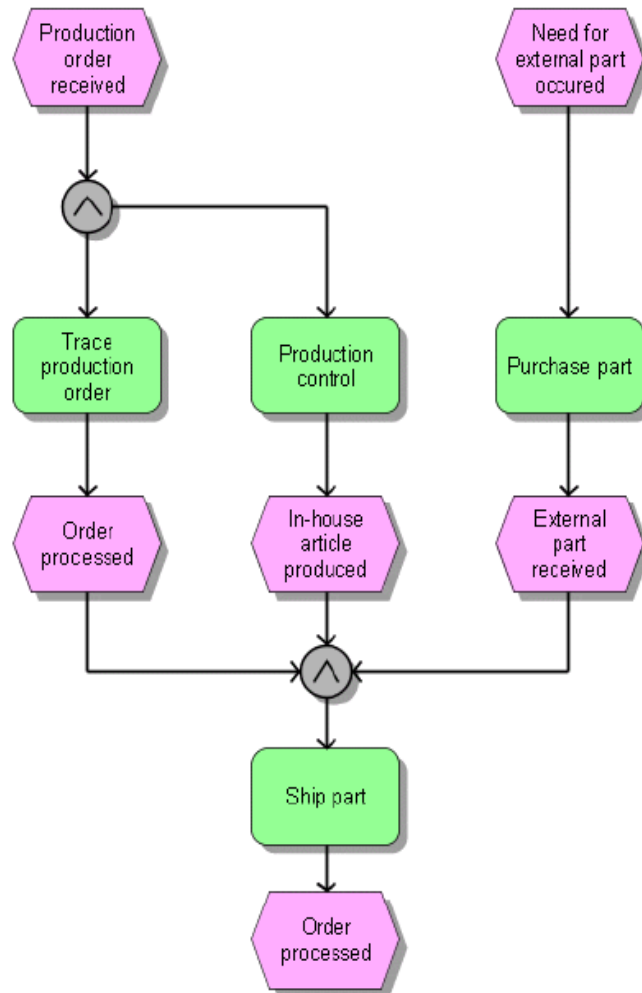
PROCESNÍ MODELOVACÍ JAZYKY

Event-driven Process Chain: EPC (1994)

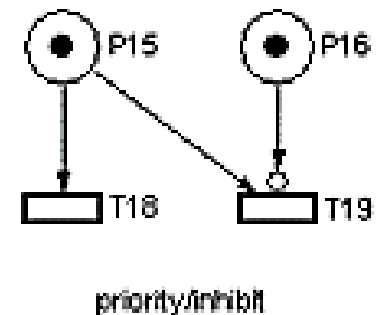
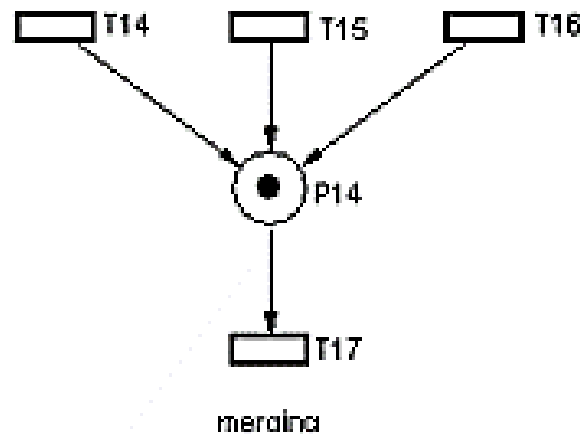
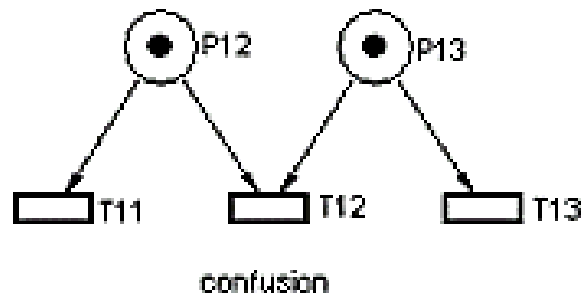
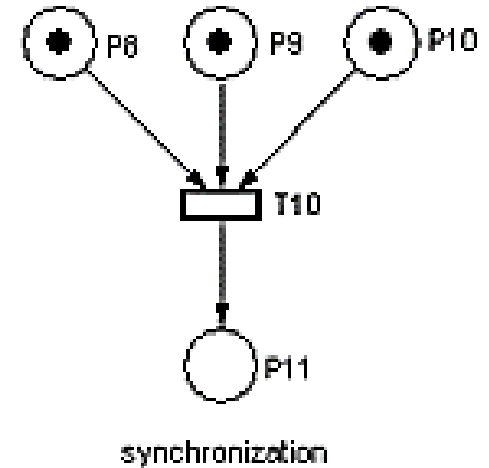
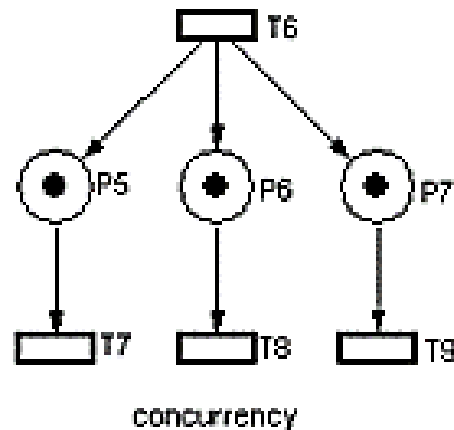
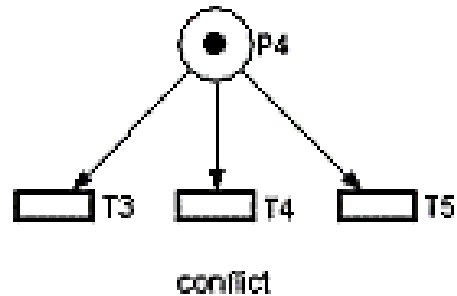
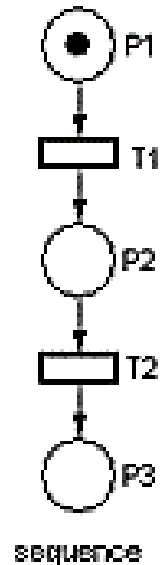


- **EPC** is conceptual business process modeling language developed in the 1990s at the Institute of Information Systems (IWi) in Saarbrücken as a result of joint project with *SAP* with a goal to define a suitable business process modeling language to document the processes of the SAP R/3 enterprise resource planning system (Mendling J. , 2008).

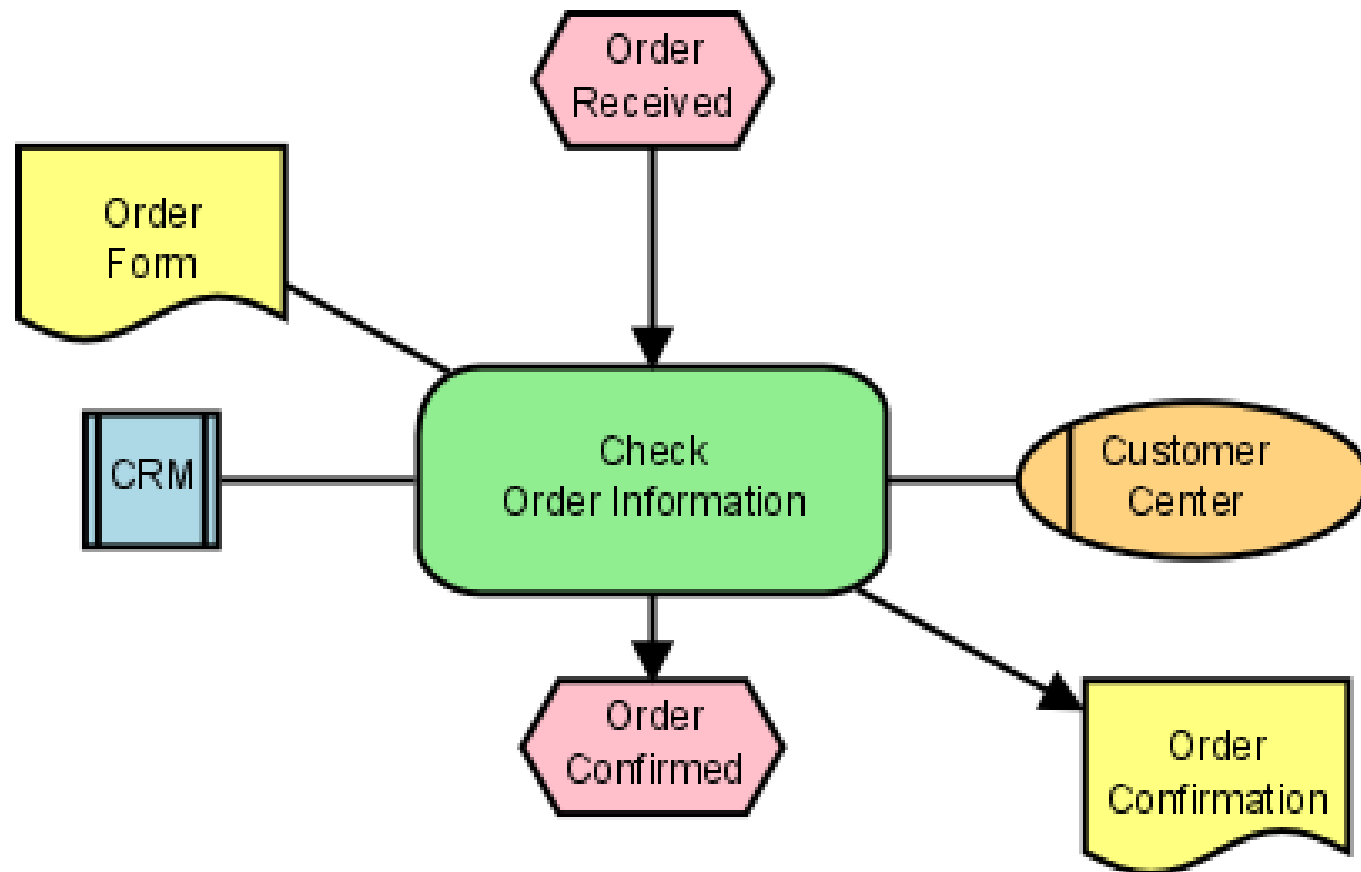
EPC

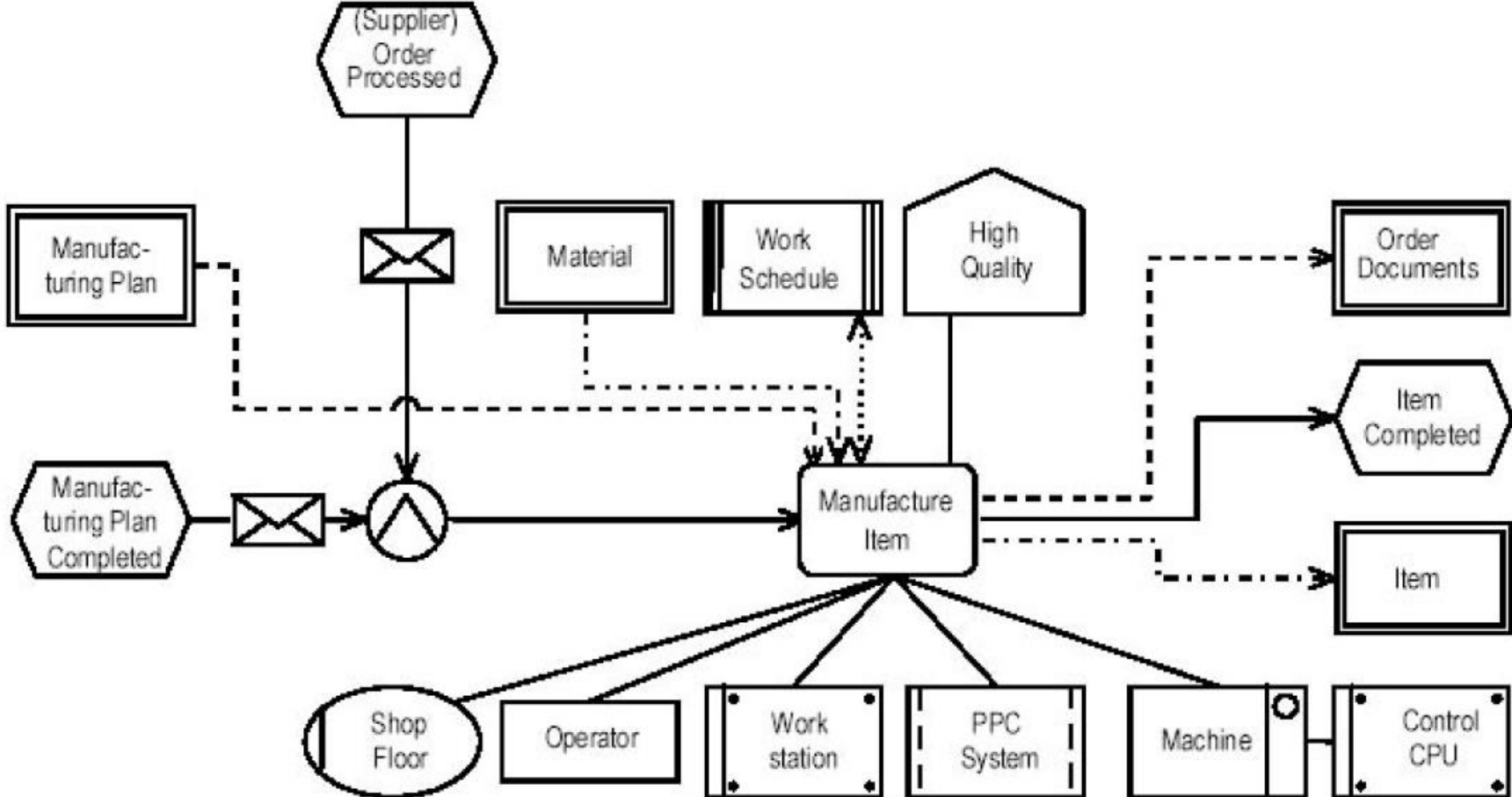


Petri Net (1962)



eEPC (ARIS)





Legend:

— Organization Flow / Resource Flow

—> Control Flow

.....> Information Flow

- - -> Information Services Flow

- . - .> Material Output Flow

PPC = Production Planning and Control Flow

Function

Event



Environmental Data

Output

Application Software

Human Output

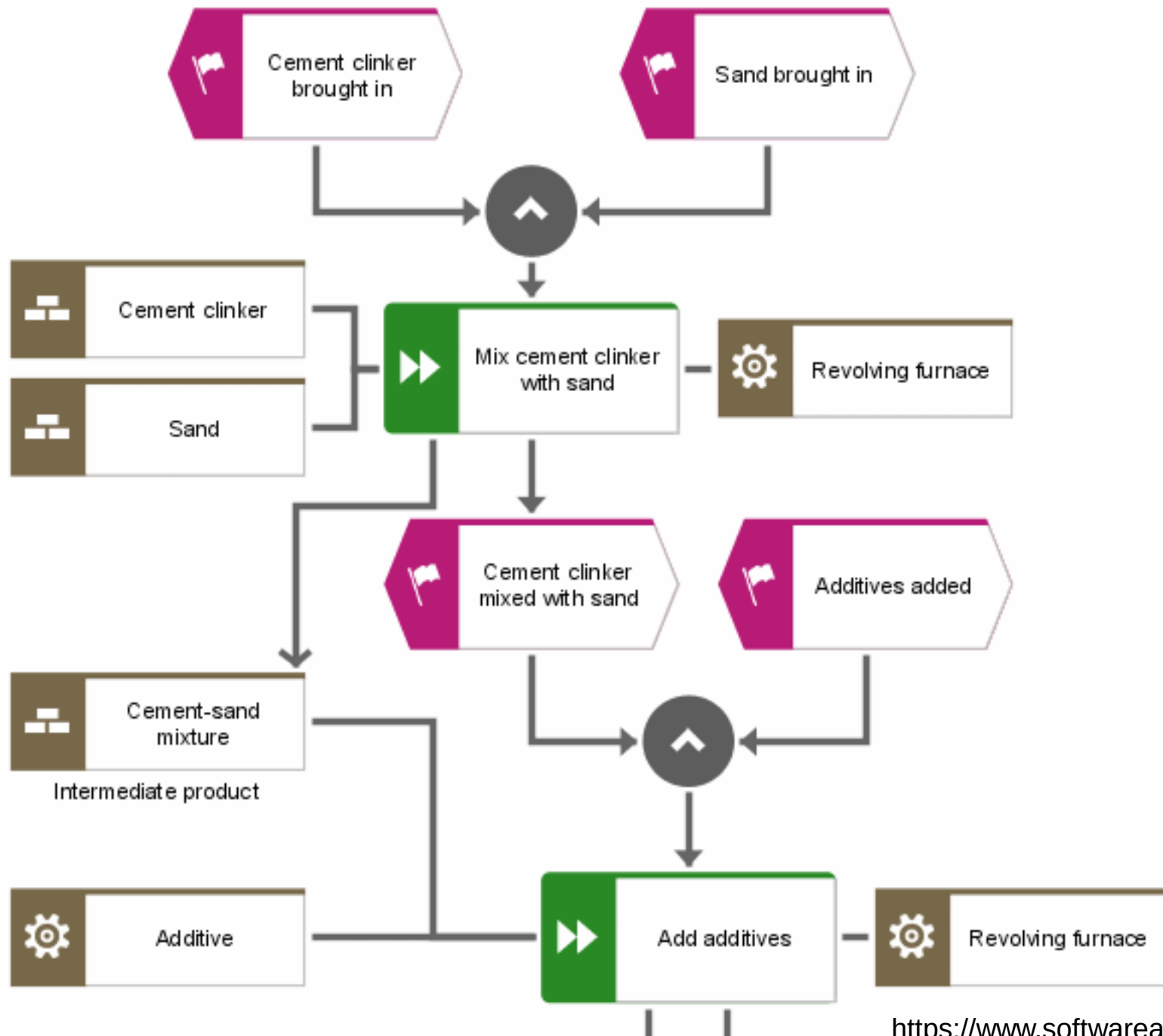
Organizational Unit

Goal

Machine

Computer Hardware





Modeling Handbook: Overview: Synchronization Naming Rules

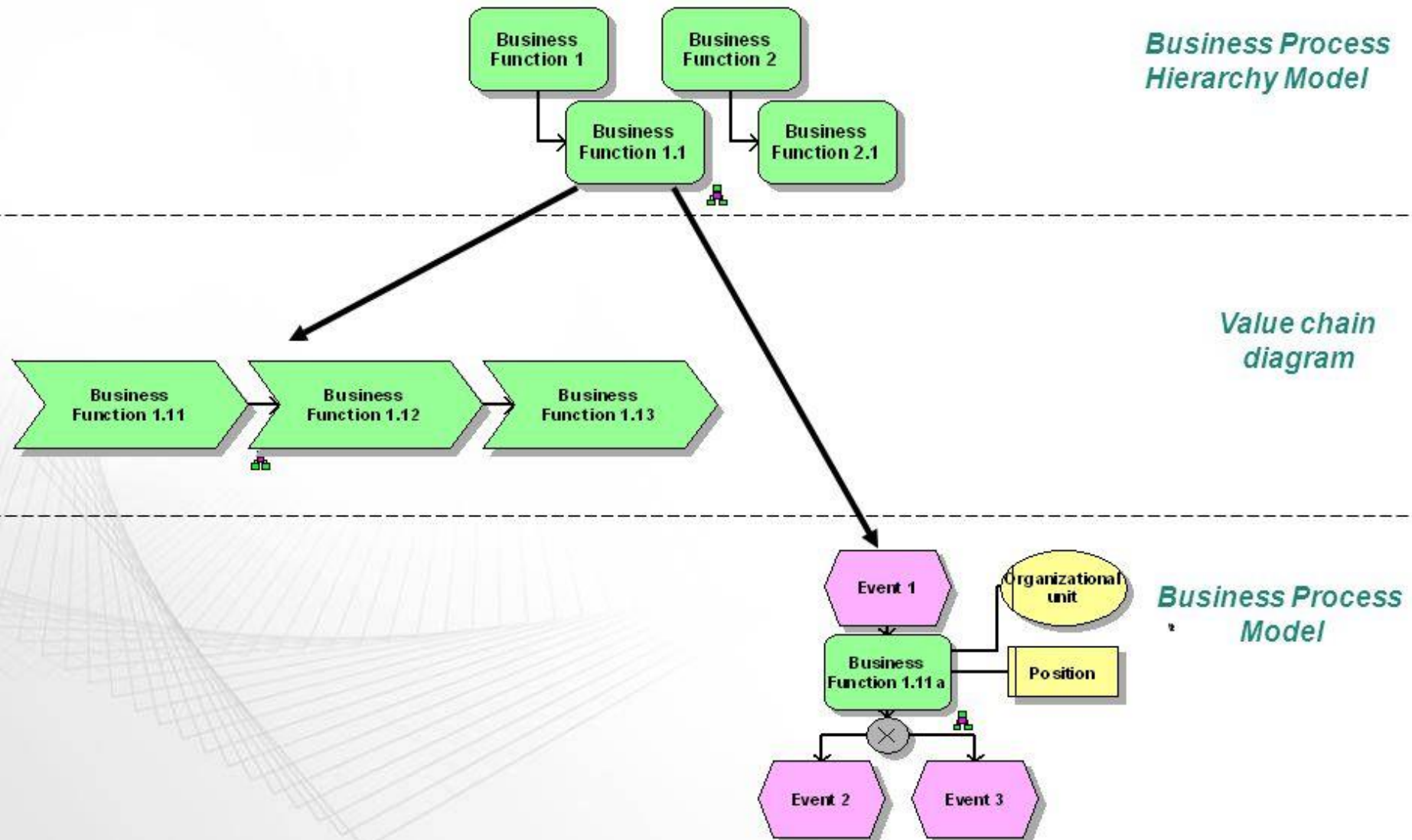


SAP Solution Manager	ARIS Model Types	SAP Field Services (Handbook)	Handbook Example	
Project	Value Added Chain Diagram	Business Process Map	Beverage	Level 0
Scenario	Value Added Chain Diagram	Business Area (Core or Support Processes)	Storage & Distribution	Level 1
	Value Added Chain Diagram	Process Group	Route Sales & DirectStore Delivery	Level 2
Business Process	Value Added Chain Diagram	Business Process	Add. Log. Services Customer Service Ex.	Level 3
	Value Added Chain Diagram	Business Process variant	Add. Log. Services Customer Service Ex.	Level 4 optional
Process Step	Event Driven Process Chain	Process Step	Create Service Notification	Level 5 = TA
	Event Driven Process Chain	Activity	Enter customer name	Level 6 = single data entry

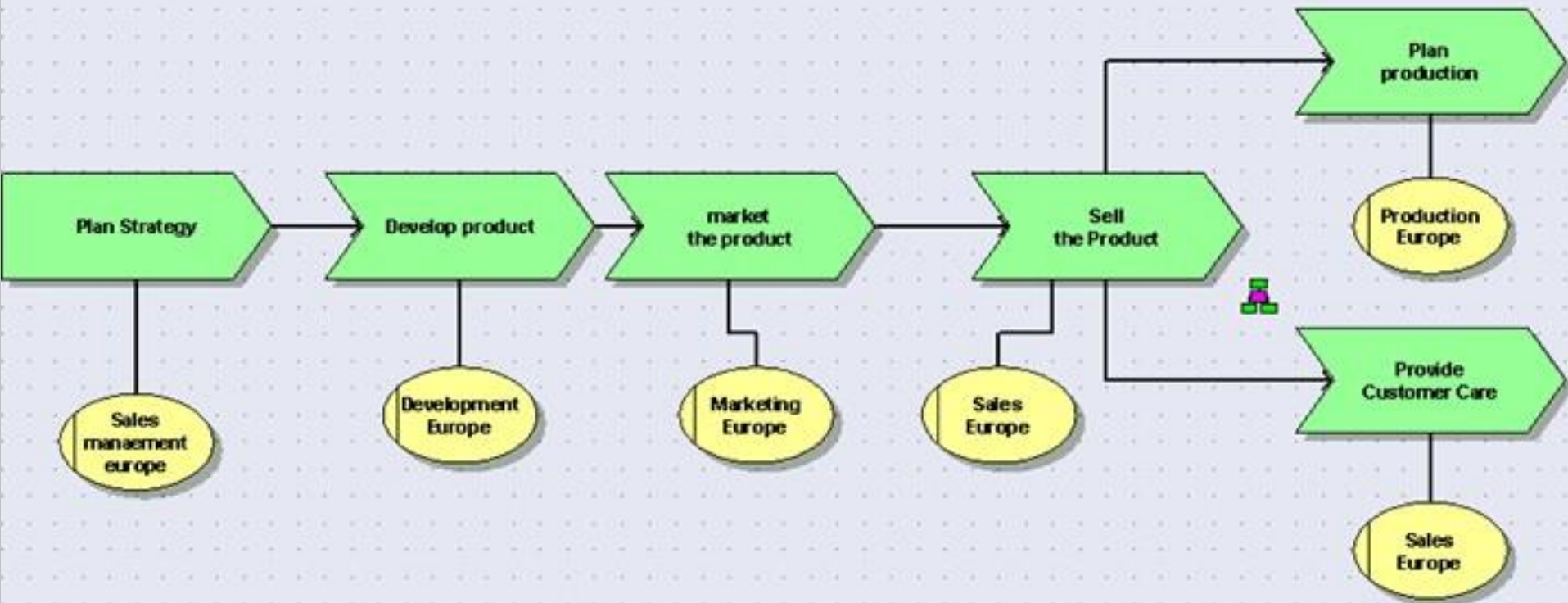
Ann Rosenberg: SAP Modeling Handbook - Modeling Standards, <https://wiki.scn.sap.com/wiki/display/ModHandbook/>

LEAD metodika – postup tvorby

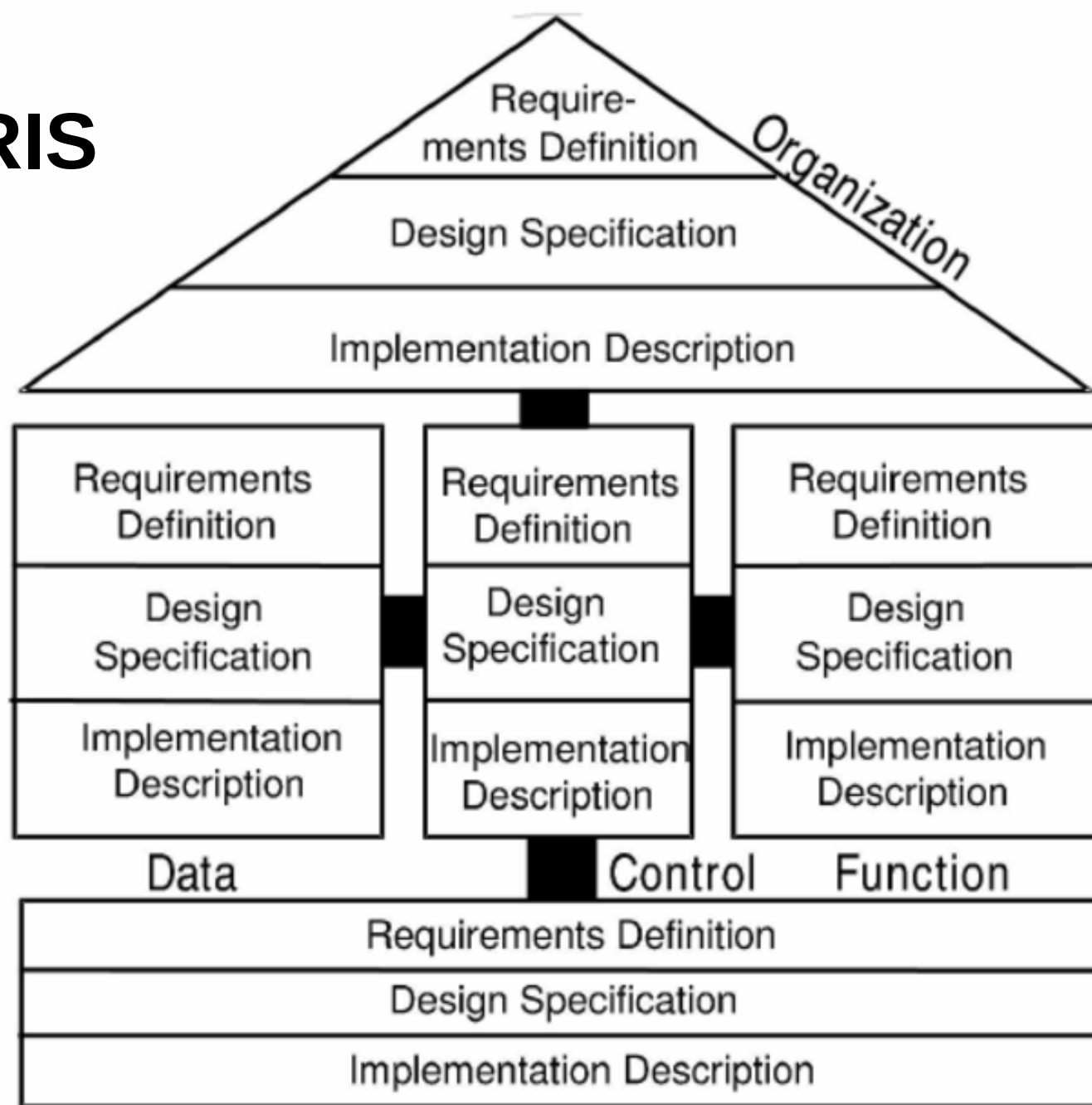
ARIS for Business Architecture



Value Chain

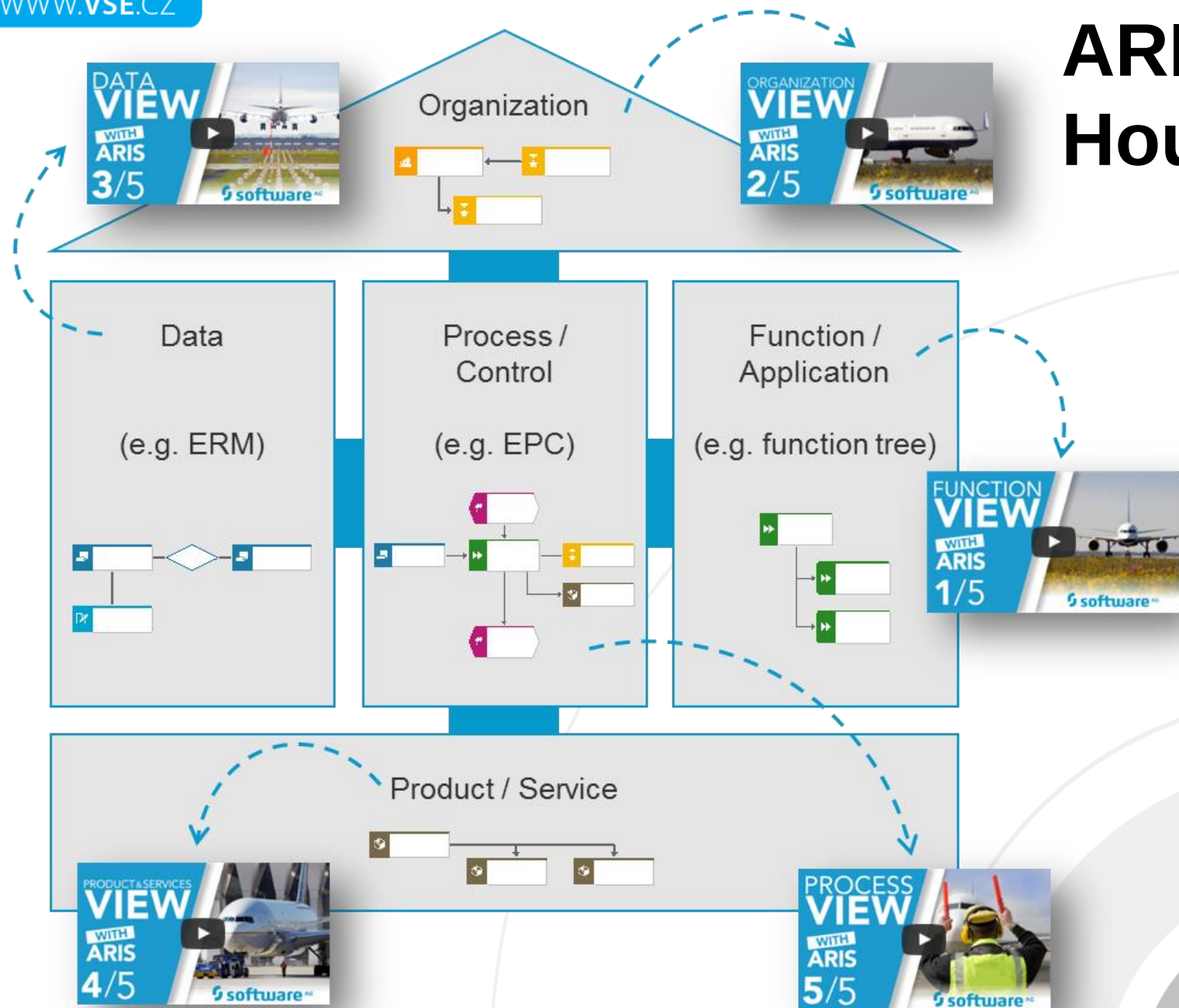


ARIS

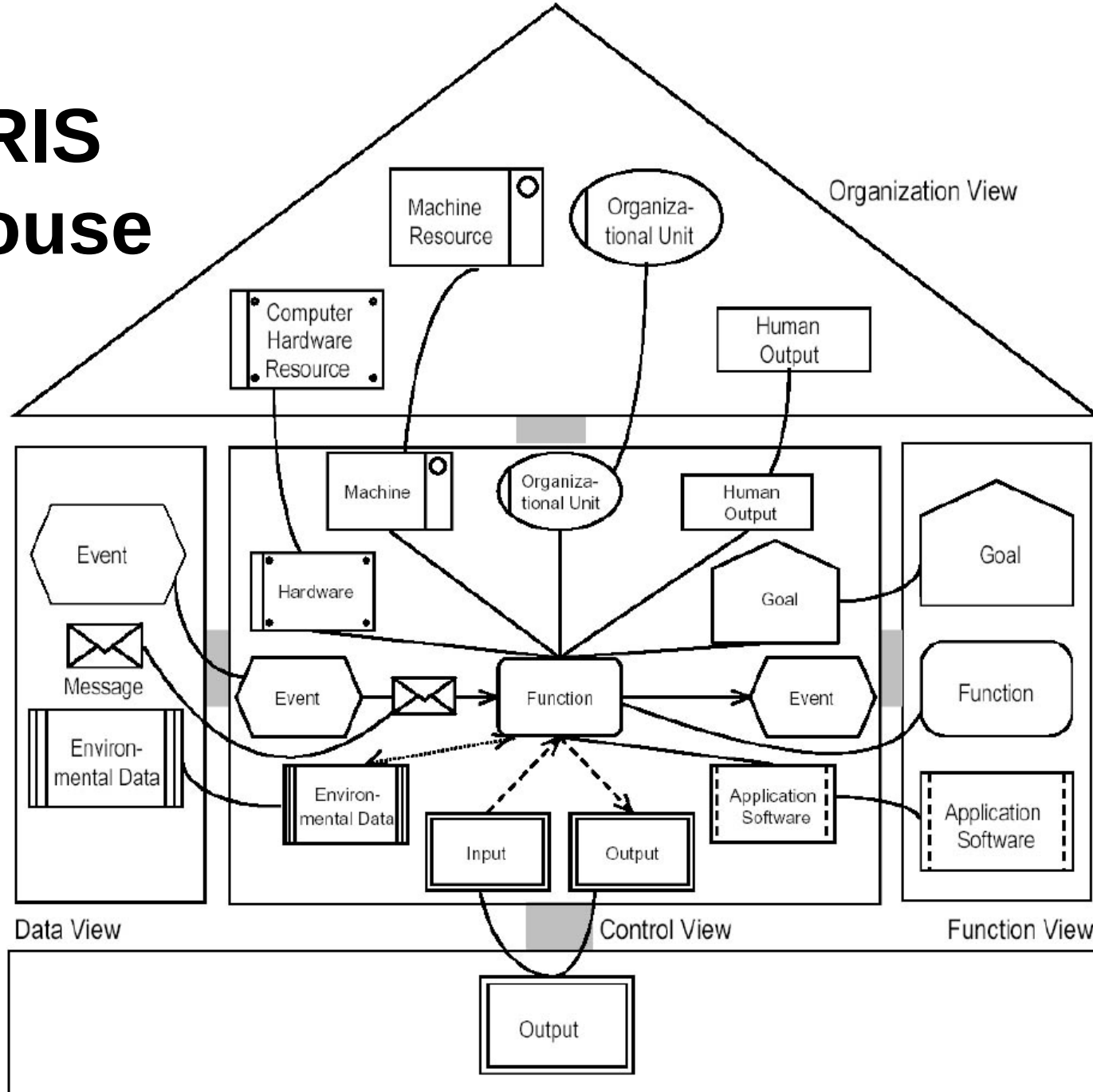


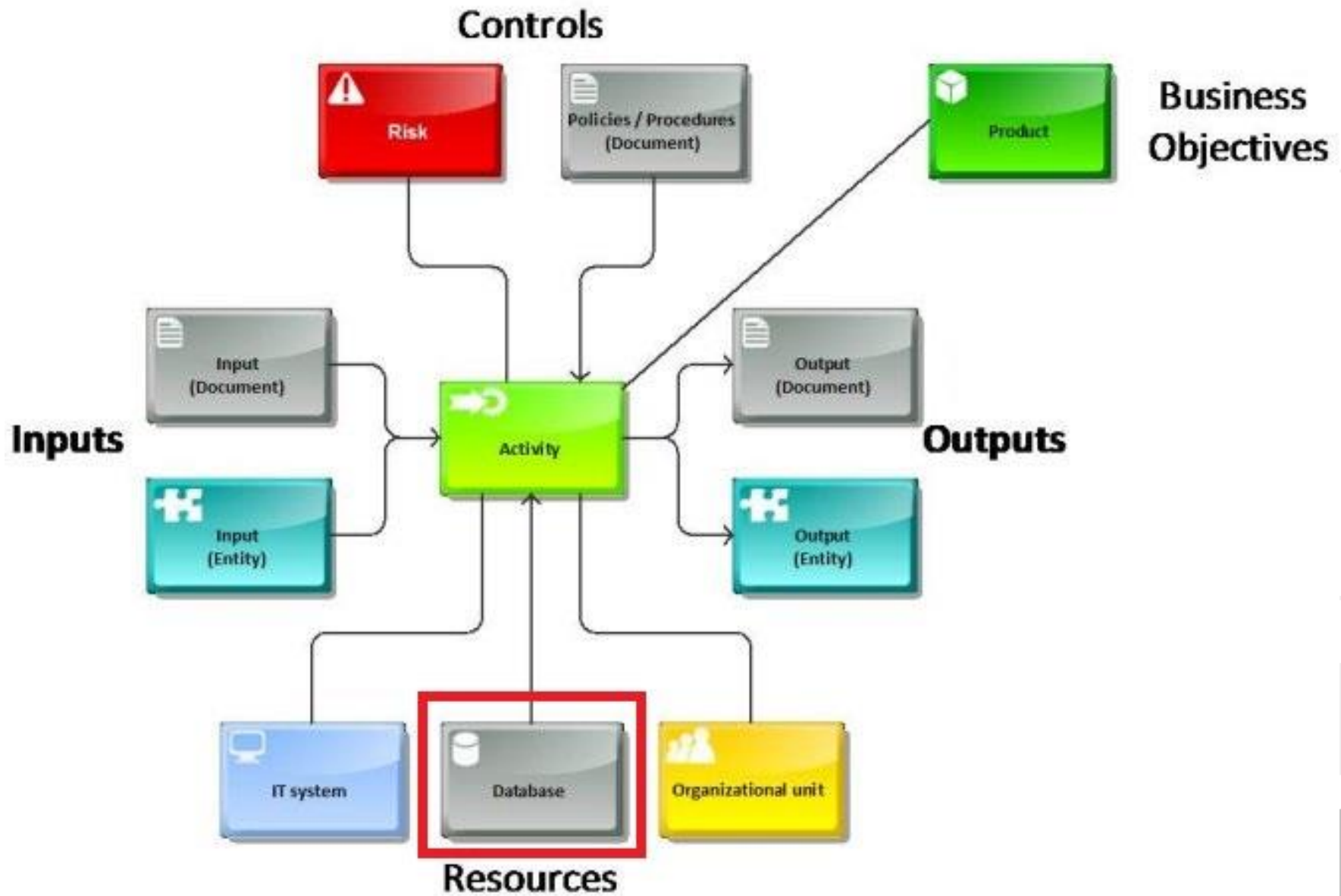
Scheer, AW. (1998). Developing the Architecture of Integrated Information Systems (ARIS House). In: ARIS — Business Process Frameworks. Springer, Berlin, Heidelberg.

ARIS House



ARIS House







Organizational chart



Network structure



Regulations



Policies



Risks



Data



Enterprise strategy



Business processes



Customer journeys



E2E scenarios



Roadmap and planning



Requirements



Capabilities



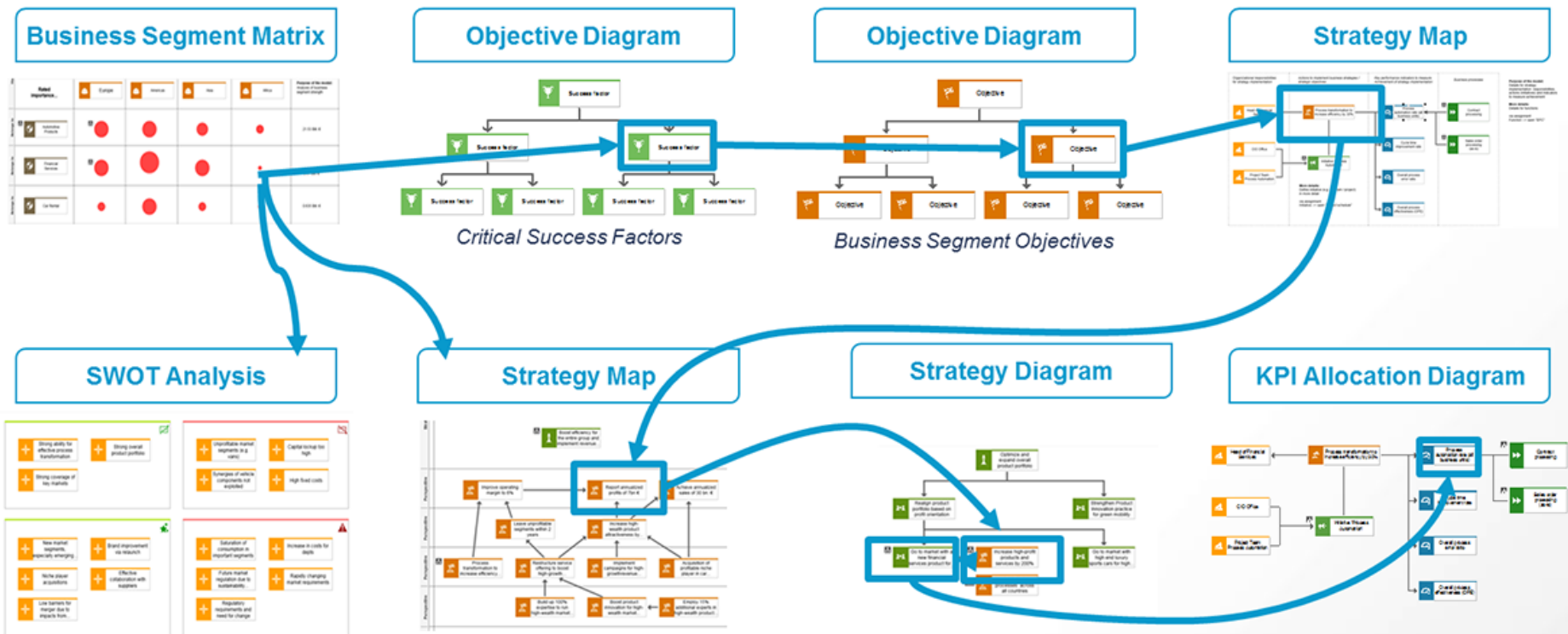
IT systems



Technologies

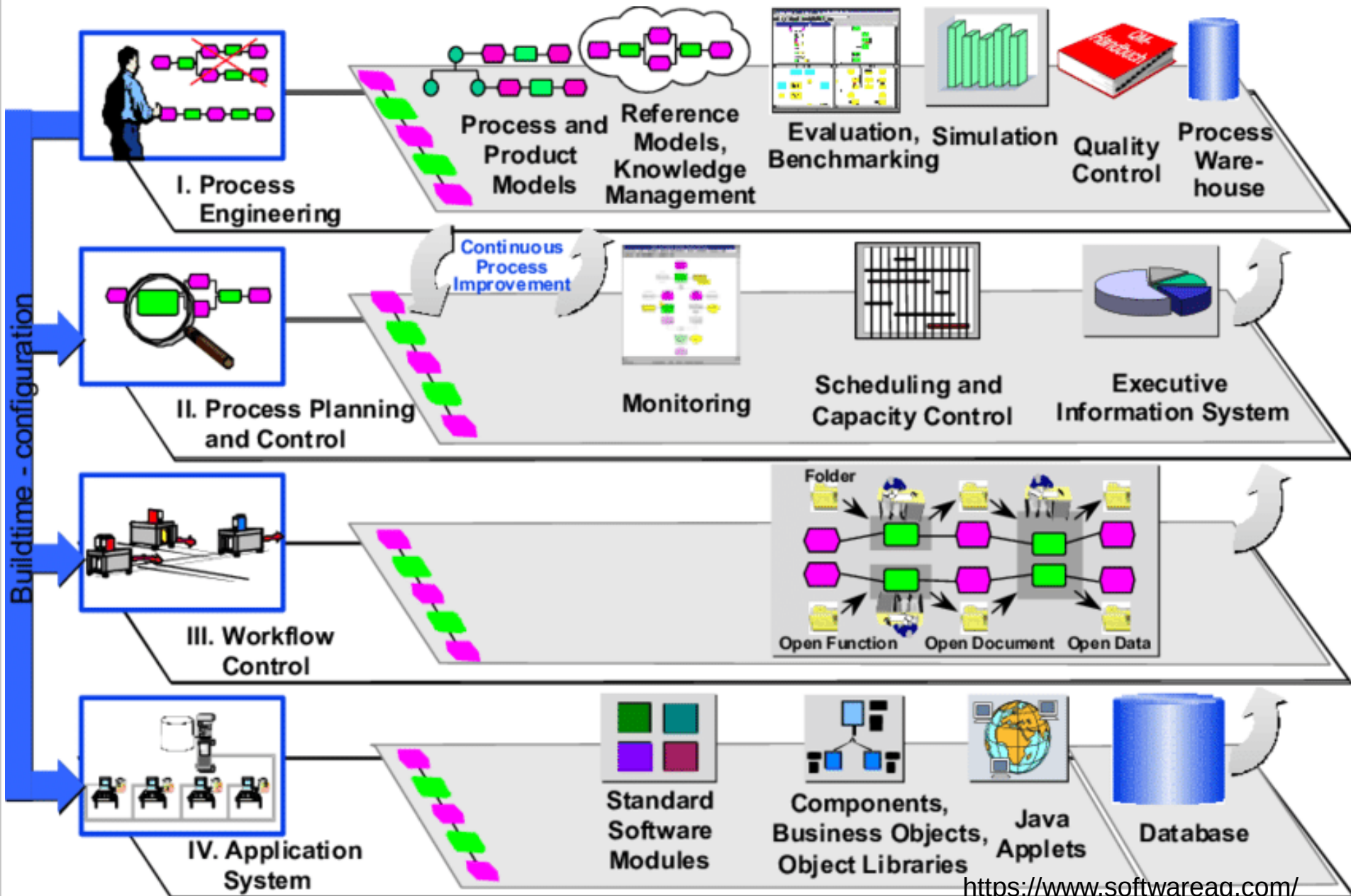


Products & Services



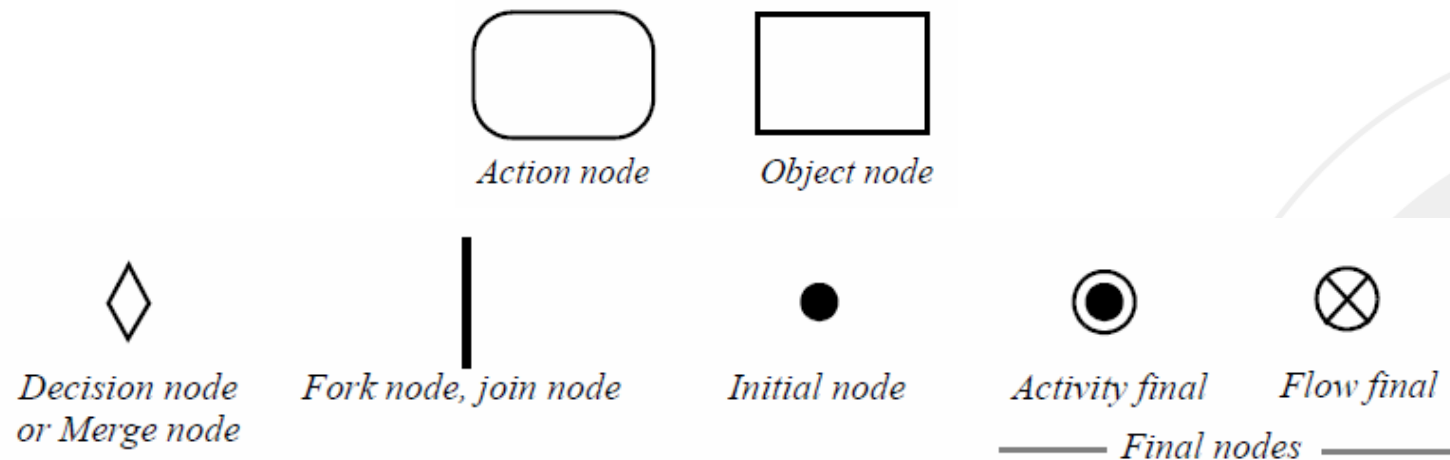
Design business models in ARIS

ARIS-House of Business Engineering

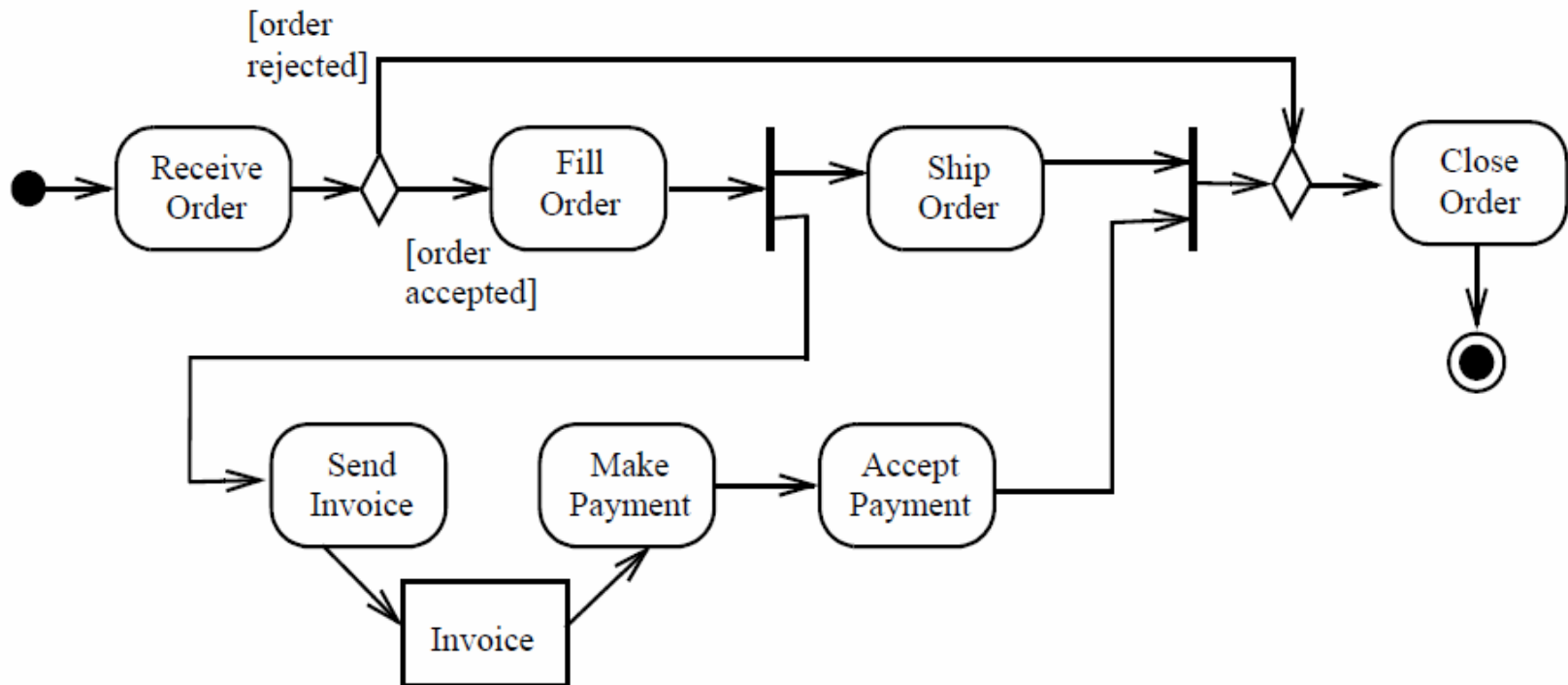


UML Activity Diagram (1997 UML 1.0)

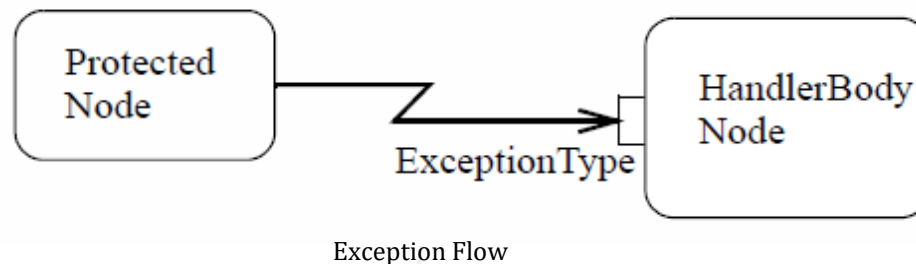
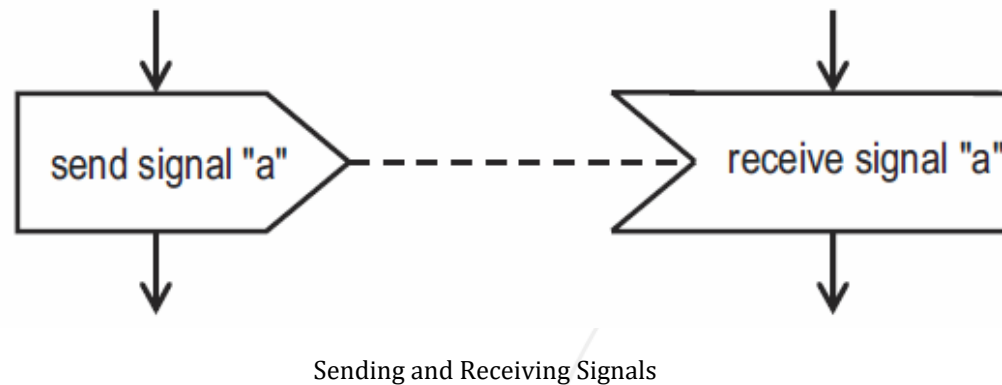
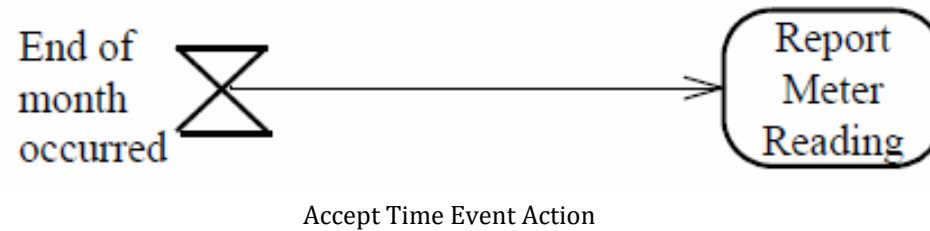
- The Unified Modeling Language (UML) is a general purpose object oriented modeling language primarily designed for modeling software systems (Engels, Förster, Heckel, & Thöne, 2005), but it can be also used for business process modeling (Object Management Group, 2011).



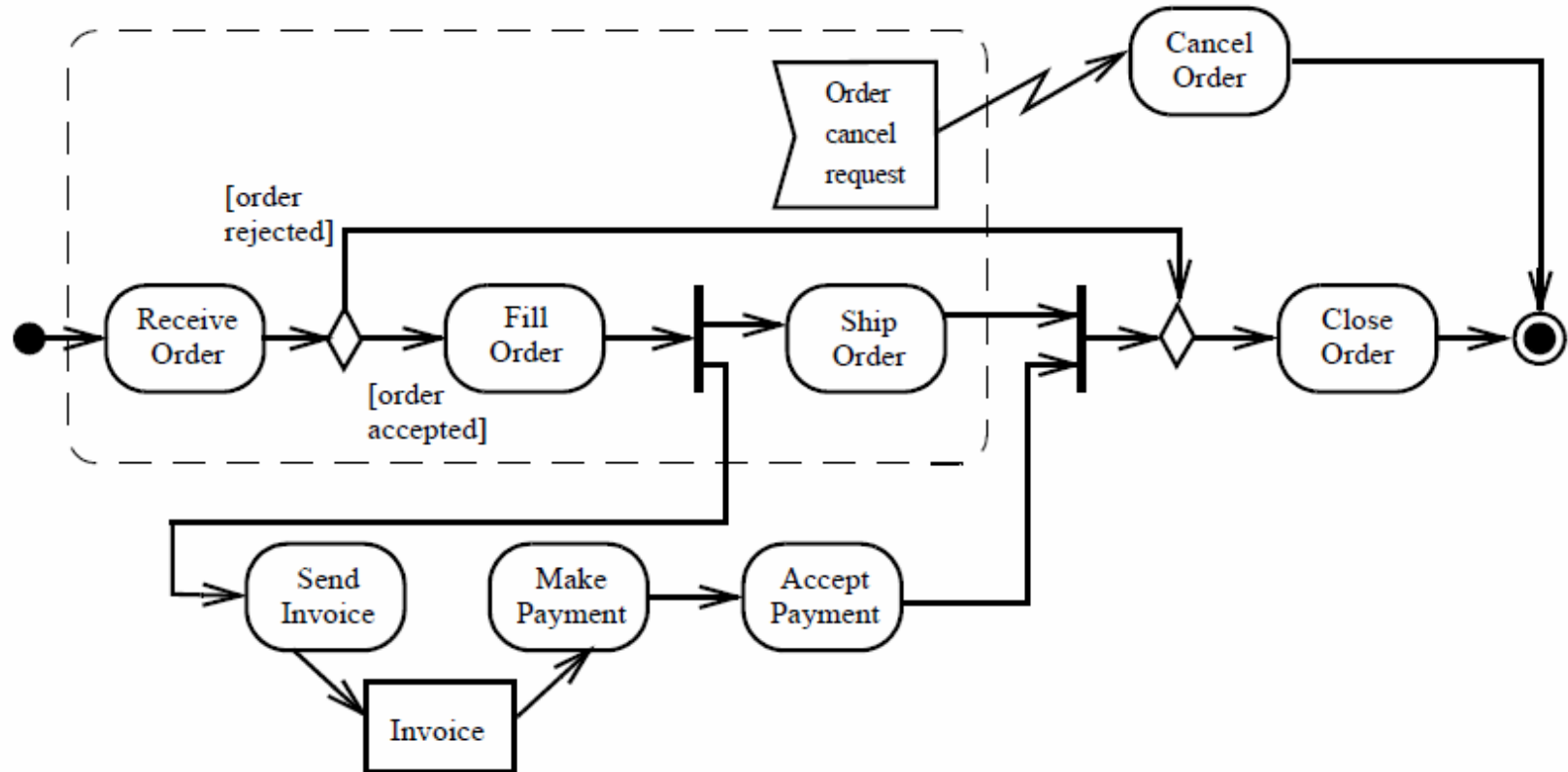
Simple Example



Extensions



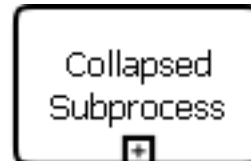
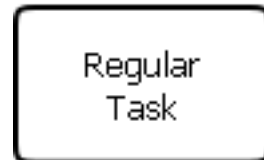
Interruption



Business Process Model and Notation - BPMN (2004)

- “The primary goal of BPMN is to provide a notation that is readily understandable by all business users, from the business analysts that create the initial drafts of the processes, to the technical developers responsible for implementing the technology that will perform those processes, and finally, to the business people who will manage and monitor those processes. Thus, BPMN creates a standardized bridge for the gap between the business process design and process implementation.
- Another goal, but no less important, is to ensure that XML languages designed for the execution of business processes, such as WSBPEL (Web Services Business Process Execution Language), can be visualized with a business-oriented notation.”

BPMN



BPMN Task Types

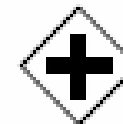
Exclusive Gateway



Event-based Gateway



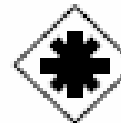
Parallel Gateway



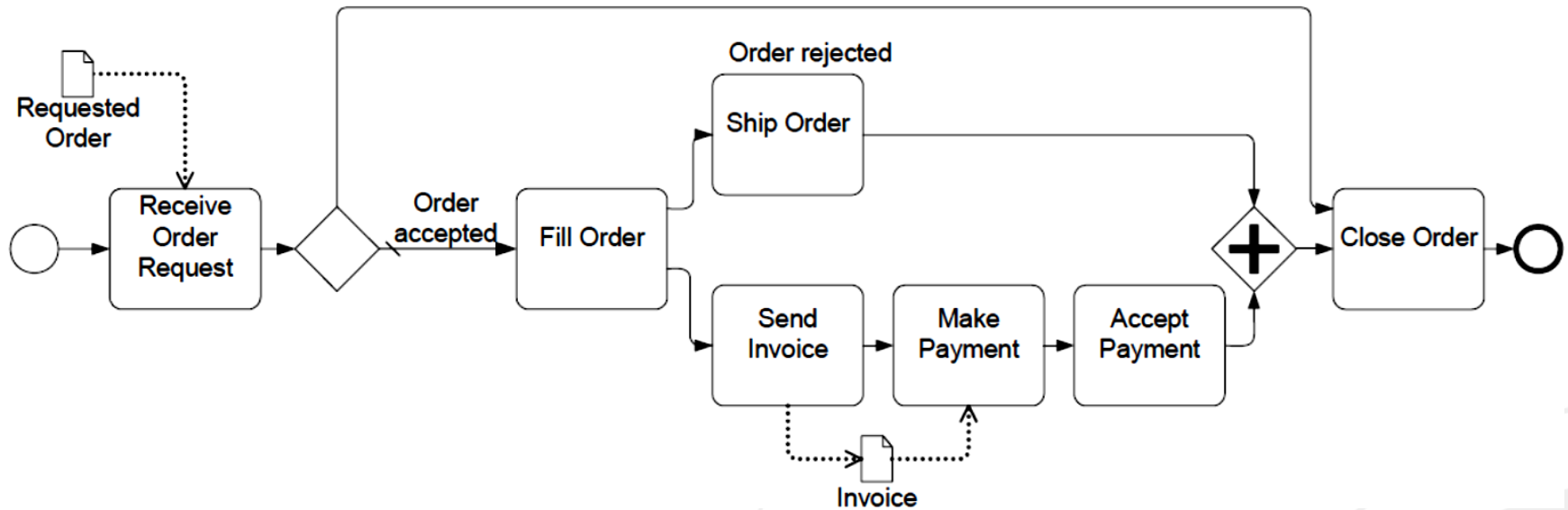
Inclusive Gateway



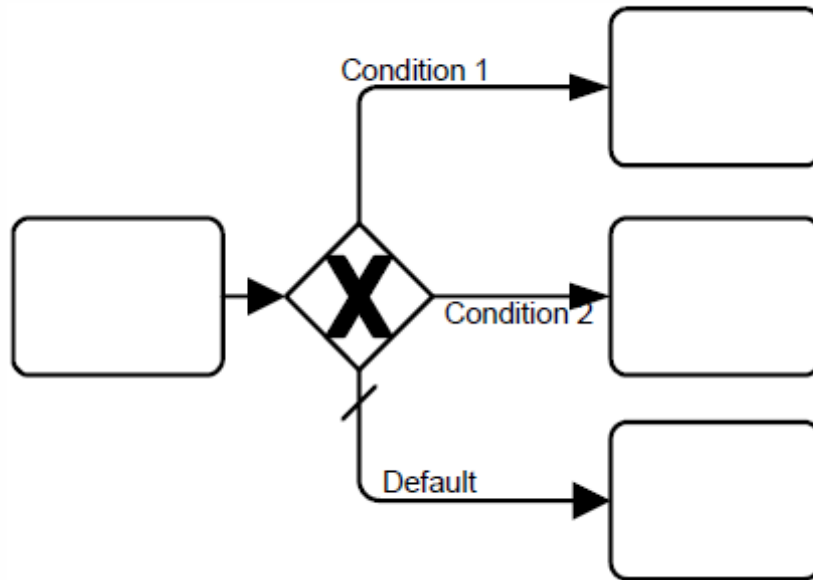
Complex Gateway



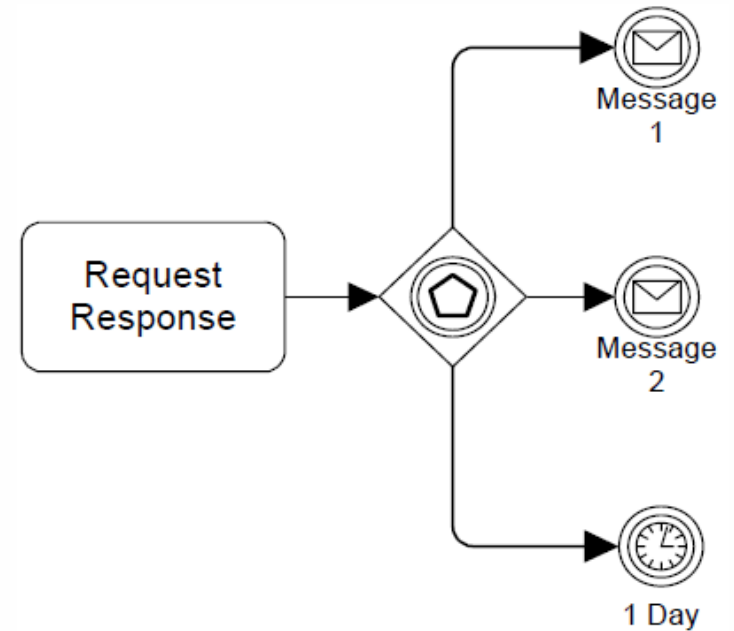
BPMN



BPMN

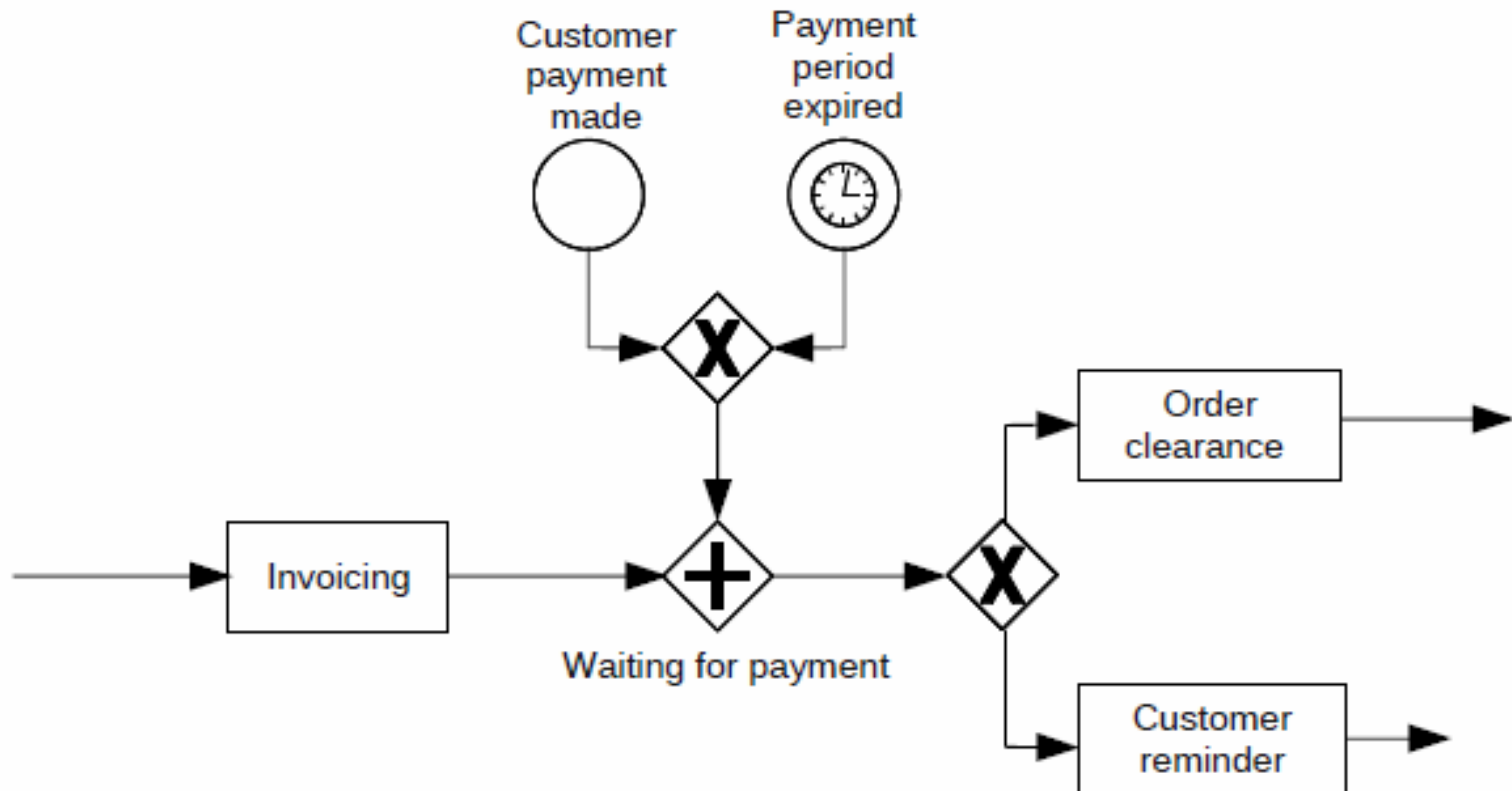


BPMN Exclusive Gateway



BPMN Event Based Gateway

Event Gateway in MMABP



	Start			Intermediate			End
	Top-Level	Event Sub-Process Interrupting	Event Sub-Process Non-Interrupting	Catching	Boundary Interrupting	Boundary Non-Interrupting	Throwing
None: Untyped events, indicate start point, state changes or final states.							
Message: Receiving and sending messages.							
Timer: Cyclic timer events, points in time, time spans or timeouts.							
Escalation: Escalating to an higher level of responsibility.							
Conditional: Reacting to changed business conditions or integrating business rules.							
Link: Off-page connectors. Two corresponding link events equal a sequence flow.							
Error: Catching or throwing named errors.							
Cancel: Reacting to cancelled transactions or triggering cancellation.							
Compensation: Handling or triggering compensation.							
Signal: Signalling across different processes. A signal thrown can be caught multiple times.							
Multiple: Catching one out of a set of events. Throwing all events defined							
Parallel Multiple: Catching all out of a set of parallel events.							
Terminate: Triggering the immediate termination of a process.							

Figure 2-13: BPMN 2.0 Events Overview (Berliner BPM-Offensive, 2010)

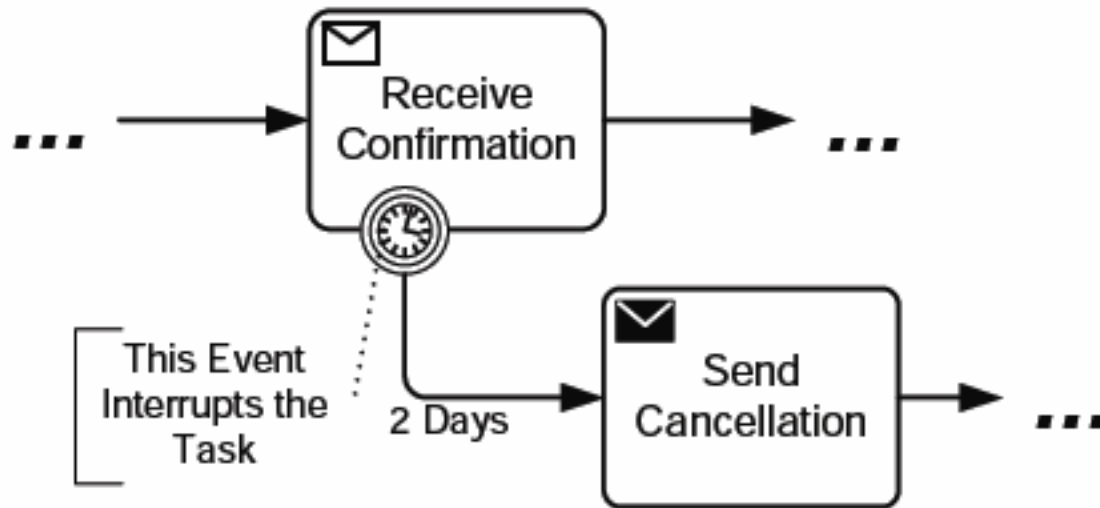


Figure 8: Use of interrupting boundary Events

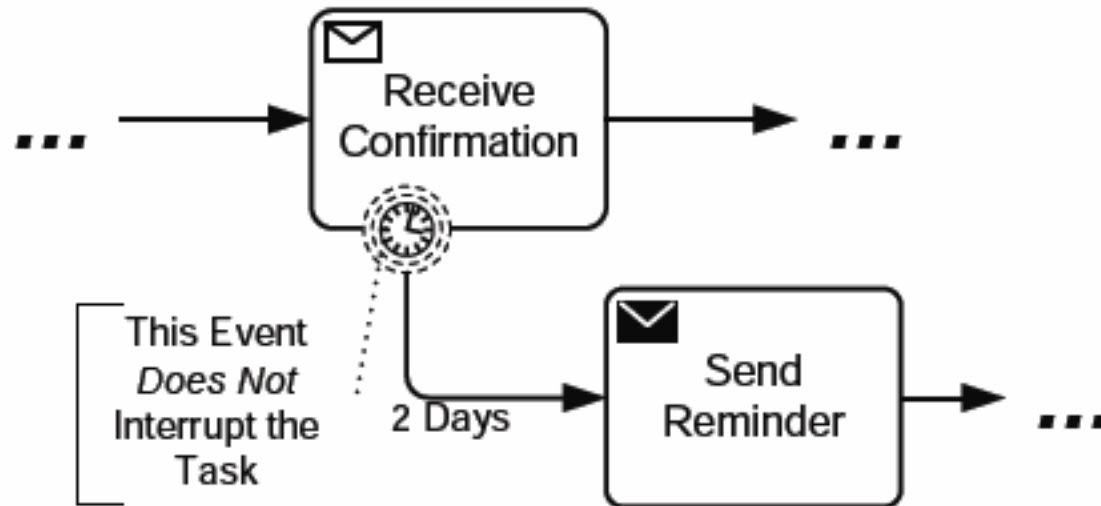
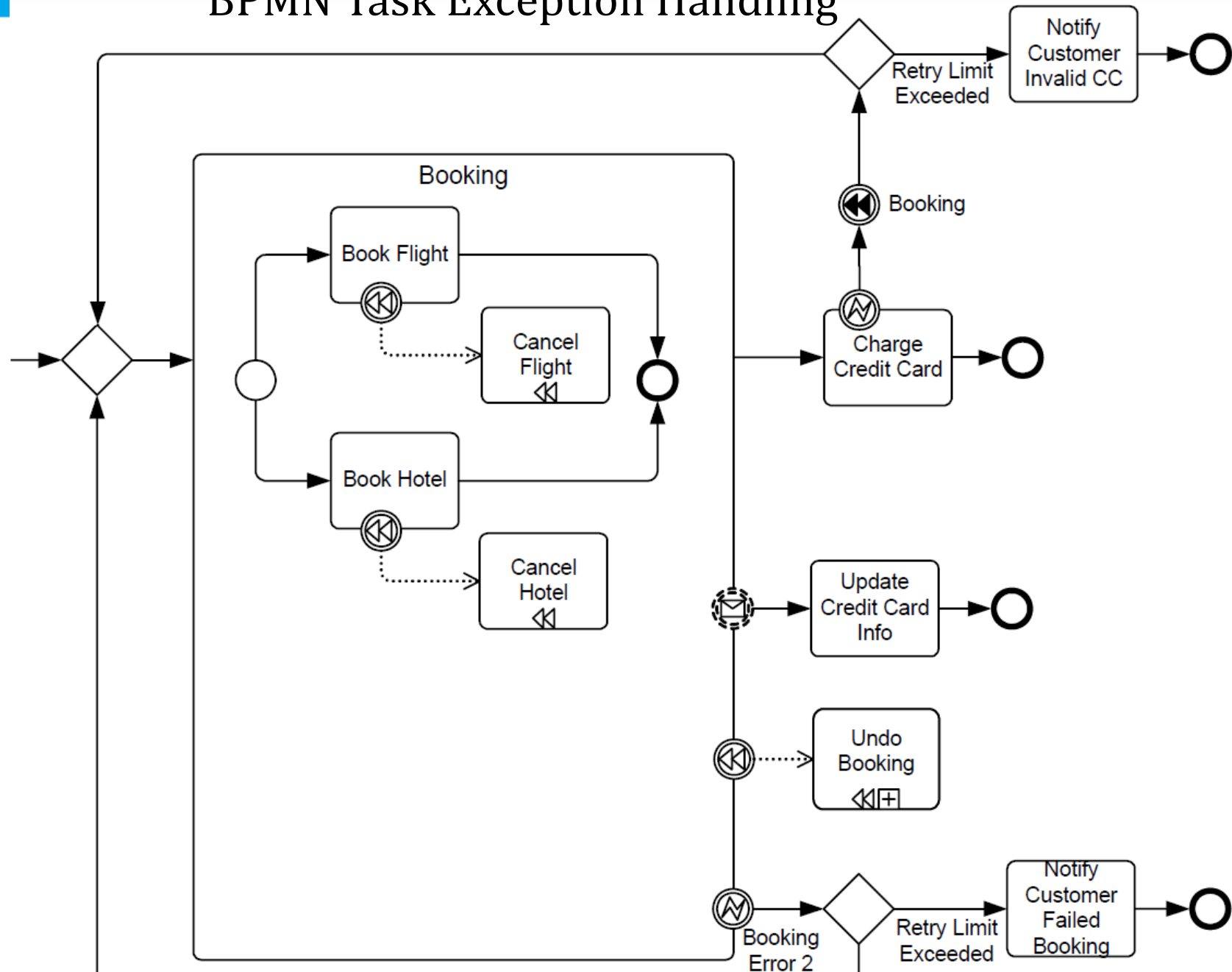


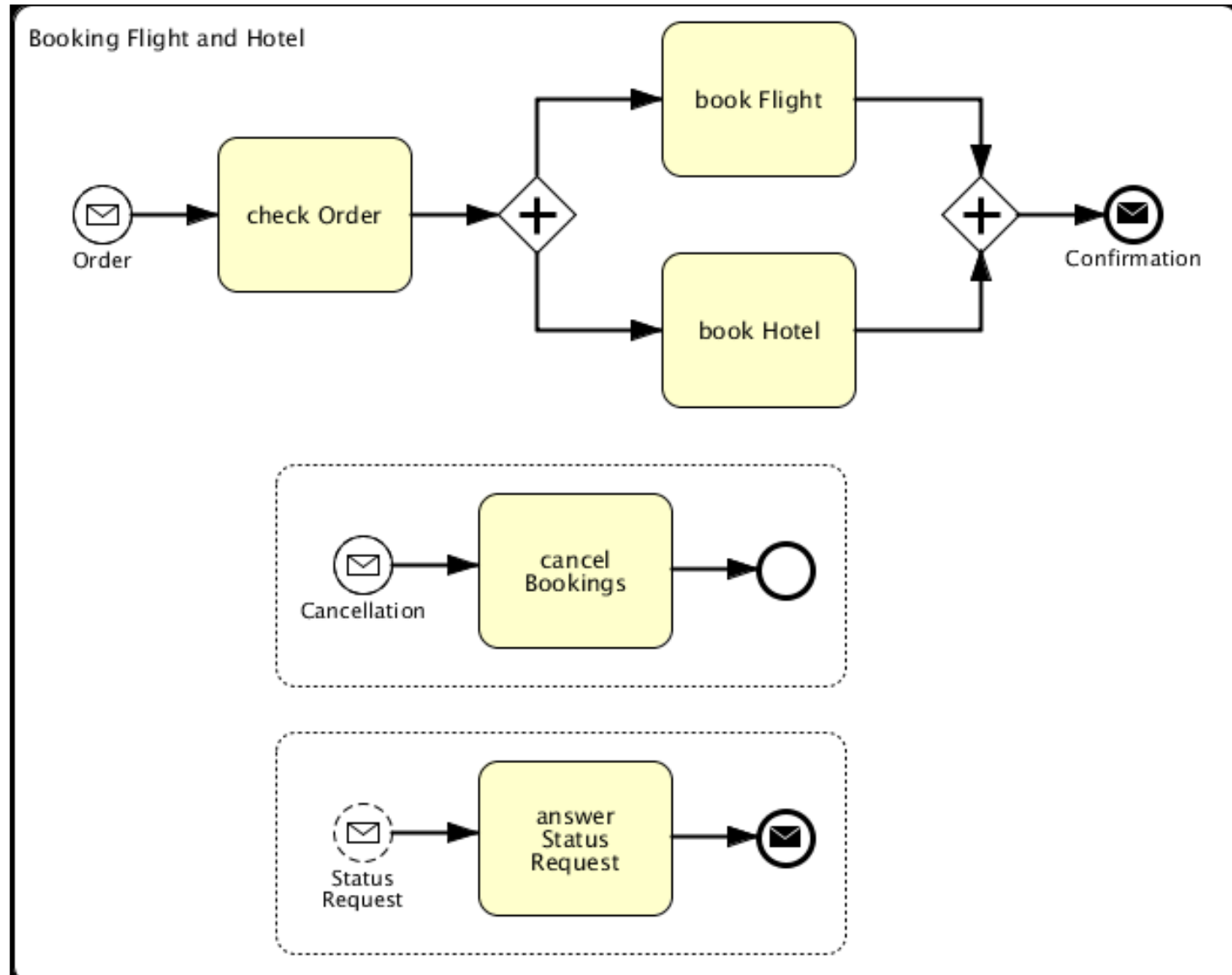
Figure 9: Use of non-interrupting boundary Events

BPMN Task Exception Handling

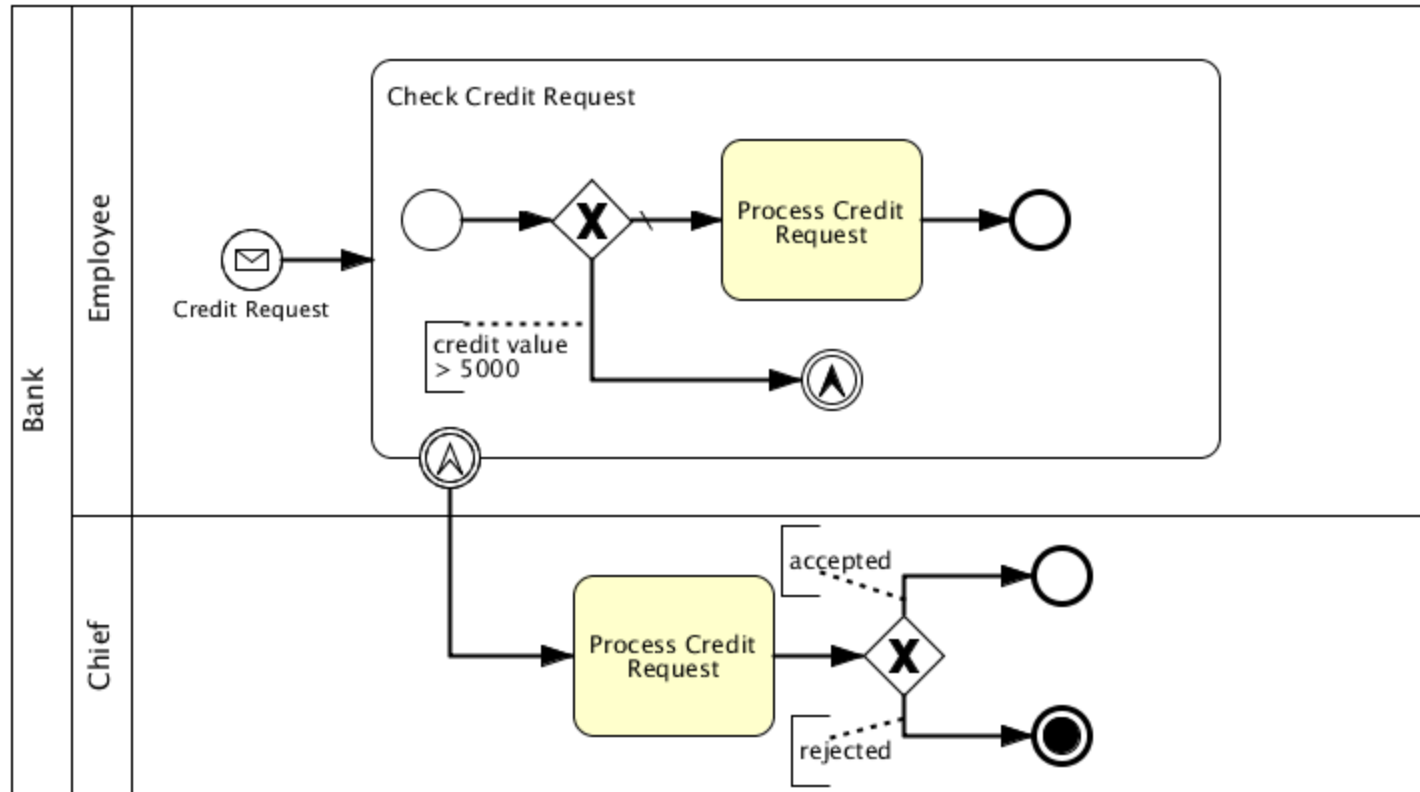


Event Sub-process

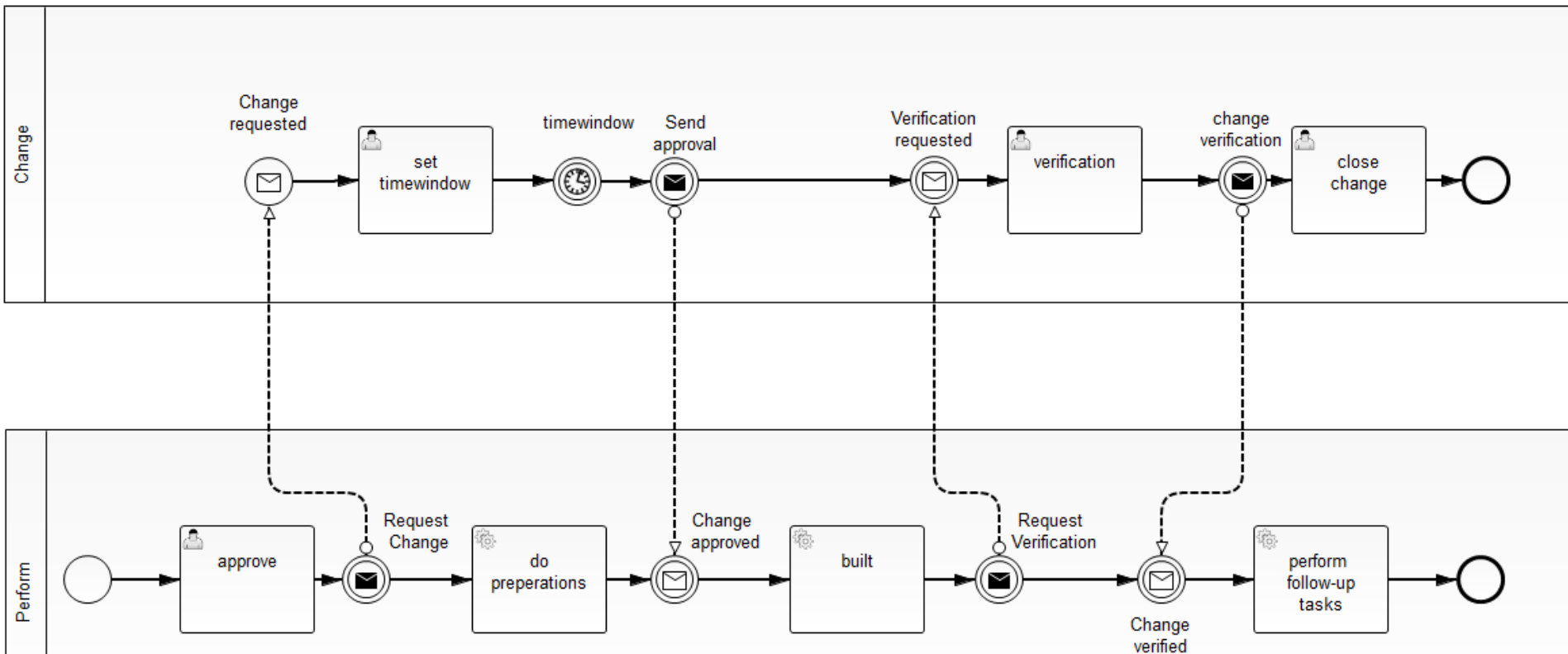
<http://en.bpmn-community.org/tutorials/6/>



Escalation



BPMN Collaboration



0%

25%

50%

75%

100%

Normal Flow

Task

End Event

Start Event / Event

Pool

Data-Based XOR

Start Message

Text Annotation

Message Flow

Parallel Fork/Join

Lanes

Gateway

Sub-Process (Collapsed)

Association

Data Object

Intermediate Timer

Intermediate Message

End Terminate

Sub-Process (Expanded)

End Link

Default Flow

Inclusive Decision/Merge

Activity Looping

Exception Task

Start Link

End Message

End Exception

Complex Decision/Merge

Event-Based XOR

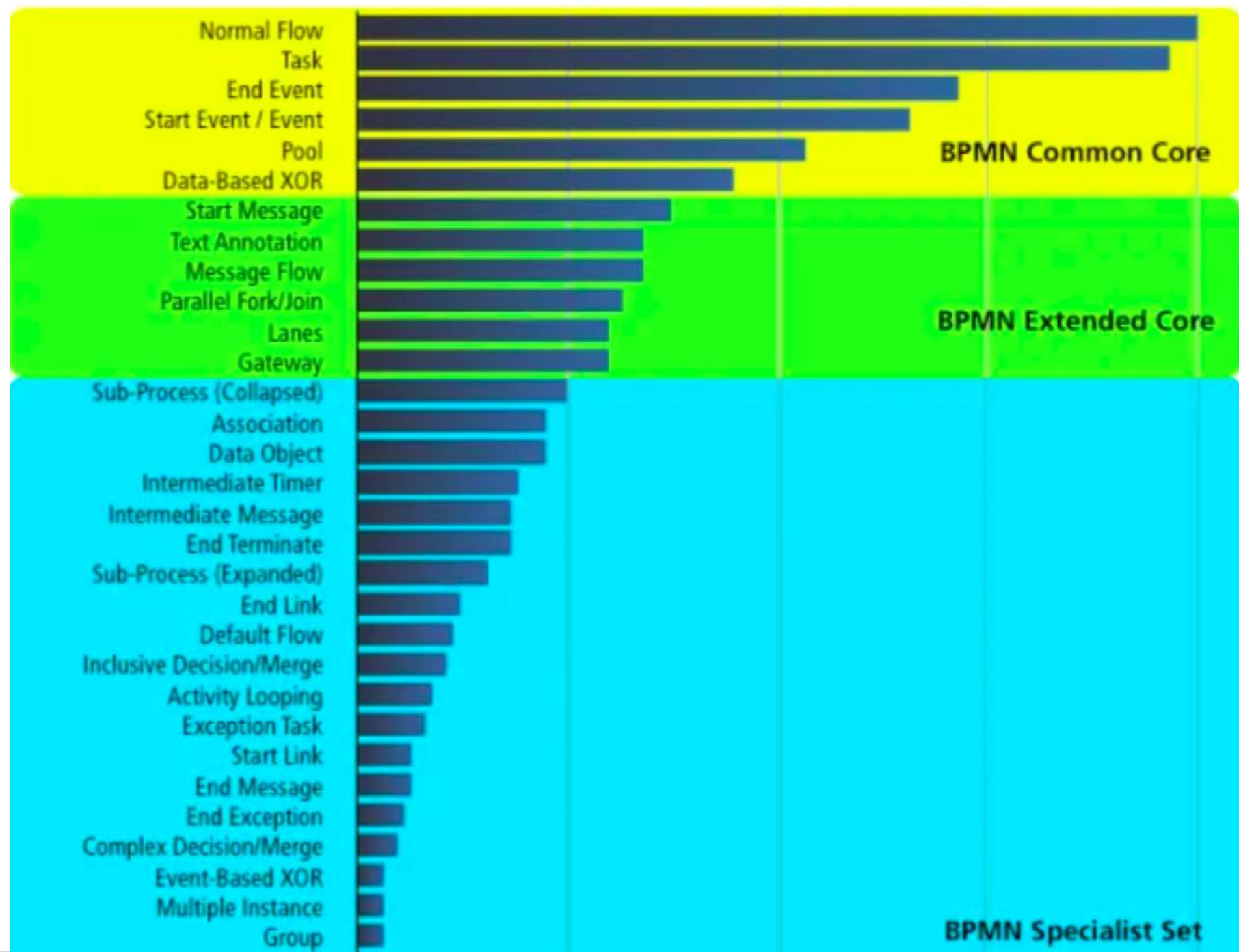
Multiple Instance

Group

BPMN Common Core

BPMN Extended Core

BPMN Specialist Set



End Cancel
Compensation
Intermediate Compensation
Conditional Flow
Exception Flow
Intermediate Link
Start Timer
Off-page connector
Start Rule
Intermediate Rule
Intermediate Multiple
End Compensation
Start Multiple
Intermediate Exception
Intermediate Cancel
End Multiple
Compensation Association

BPMN Overhead

BPMN

- Focus on executability (BPEL)
- Detailed events
- Service Oriented Architecture (SOA)
 - Making part of the functional code readable for the business users

Decision Model and Notation (DMN)

- Industry standard for modeling and executing decisions that are determined by business rules.
- Common notation that is understandable by all business users, from the business analysts to the technical developers

Rozhodovací tabulka

Rozhodnutí, co nabídnete hostům pozvaných na večeři

Dish

[Show details](#)

U	Input +		Output +	Annotation 
	Season		Dish	
1	"Fall"		"Spareribs"	-
2	"Winter"		"Roastbeef"	-
3	"Spring"		"Steak"	-
4	"Summer"		"Light Salad and a nice Steak"	Hey, why not!?
+	-		-	-

Dish

[Show details](#)

U	Input +		Output +	Annotation 
	Season	Vegetarian Guests	Dish	
1	"Fall"	false	"Spareribs"	-
2	"Winter"	false	"Roastbeef"	-
3	"Spring"	false	"Steak"	-
4	"Summer"	false	"Light Salad and a nice Steak"	Hey, why not!?
5	-	true	"Pasta"	-
+	-	-	-	-

DMN Decision Table

<https://camunda.com/dmn/>

Decision Name & Id

Hit Policy

Input Expression

Input Type Definition

Output Name

Output Type Definition

Hide details

Dish decision				
U	Input +		Output +	 Annotation
	Season	How many guests	Dish	
	season	guestCount	desiredDish	
	string	integer	string	
1	"Fall"	<= 8	"Spareribs"	-
2	"Winter"	<= 8	"Roastbeef"	-
3	"Spring"	<= 4	"Dry Aged Gourmet Steak"	-
4	"Spring"	[5..8]	"Steak"	Save money
5	"Fall", "Winter", "Spring"	> 8	"Stew"	Less effort
6	"Summer"	-	"Light Salad ad nice Steak"	Hey, why not?
+	-	-	-	-

Input Entry (Condition)

Rule

Output Entry (Conclusion)

Friendly Enough Expression Language (FEEL).

- FEEL defines a syntax for expressing conditions that input data should be evaluated against.

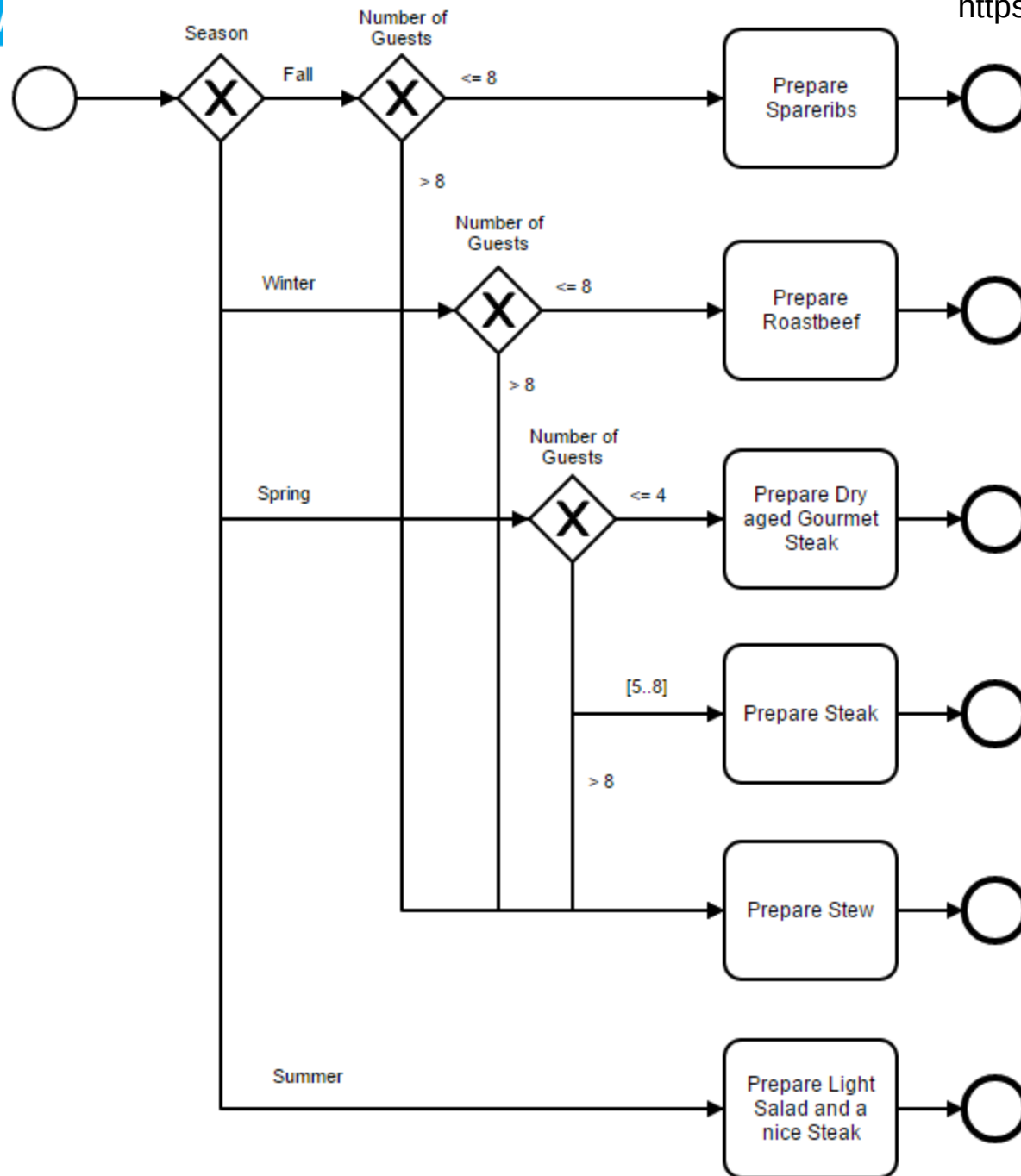
[Hide details](#)

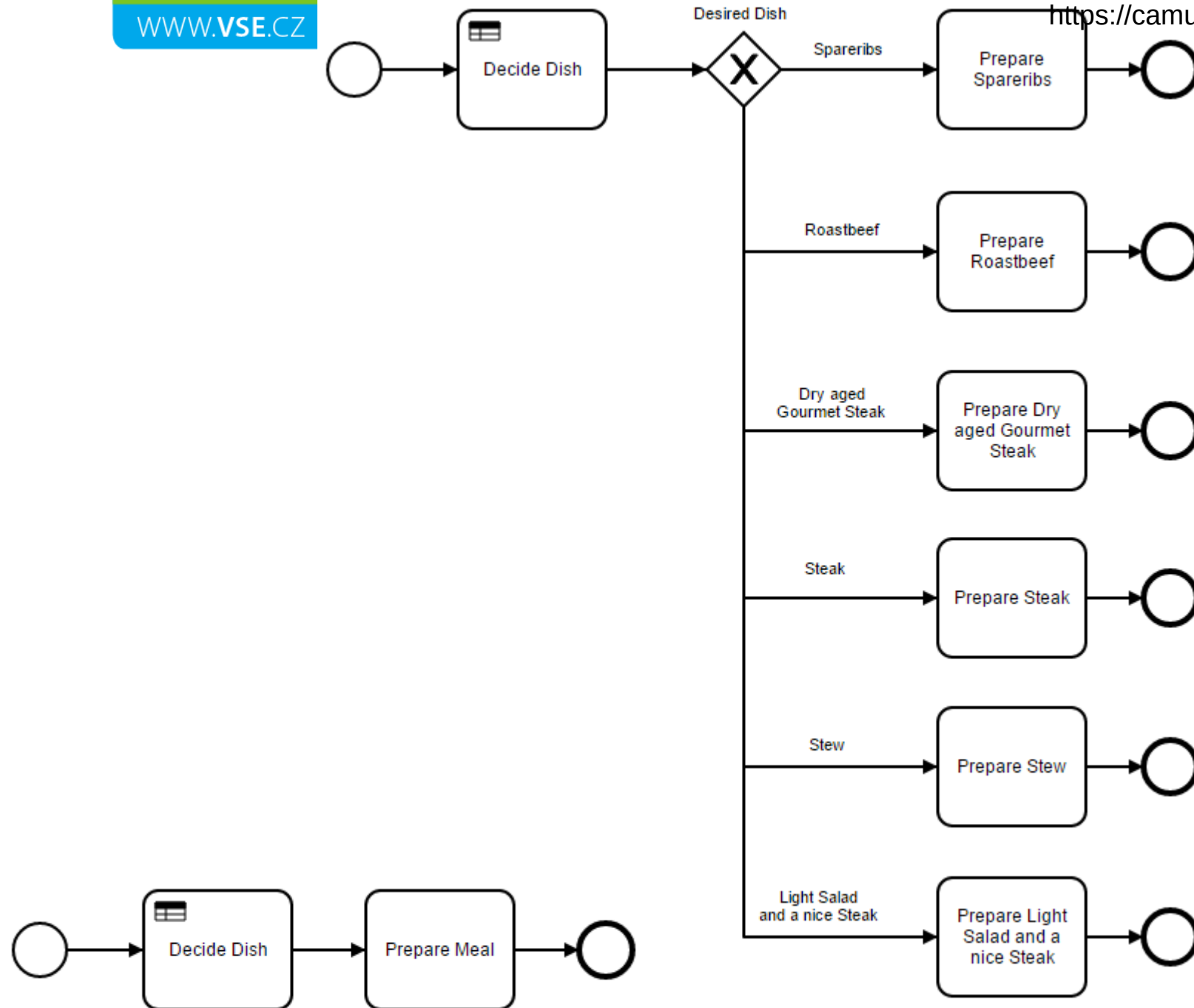
Dish

decision



U	Input +		Output +	Annotation
	Season	How many guests	Dish	
	season	guestCount	desiredDish	
	string	integer	string	
1	"Fall"	<= 8	"Spareribs"	-
2	"Winter"	<= 8	"Roastbeef"	-
3	"Spring"	<= 4	"Dry Aged Gourmet Steak"	-
4	"Spring"	[5..8]	"Steak"	Save money
5	"Fall", "Winter", "Spring"	> 8	"Stew"	Less effort
6	"Summer"	-	"Light Salad and a nice Steak"	Hey, why not!?
+	-	-	-	-





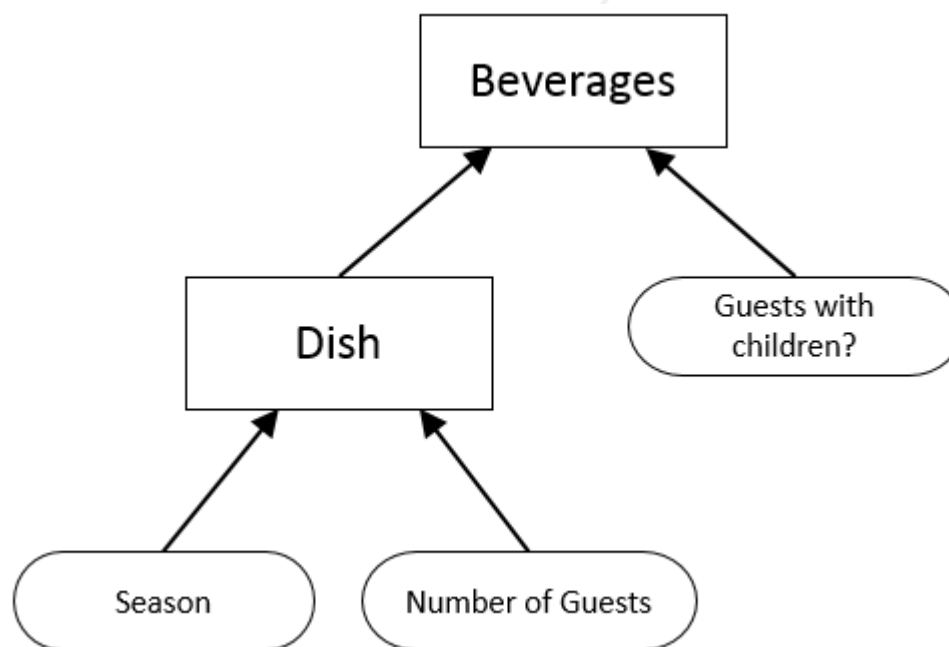
Decision Requirements Diagrams

- Complex decisions, that may be composed of other decisions
- Decisions: The act of determining an output value from a number of input values, using logic definition. This decision logic is what you can express in decision tables.
- Input Data: The input data that you "feed" into your decision logic in order to determine the output value.
- Relations between Decisions: You can connect decisions with arrows and therefore indicate which decision output will be considered as an input for another decision.

[Show details](#)

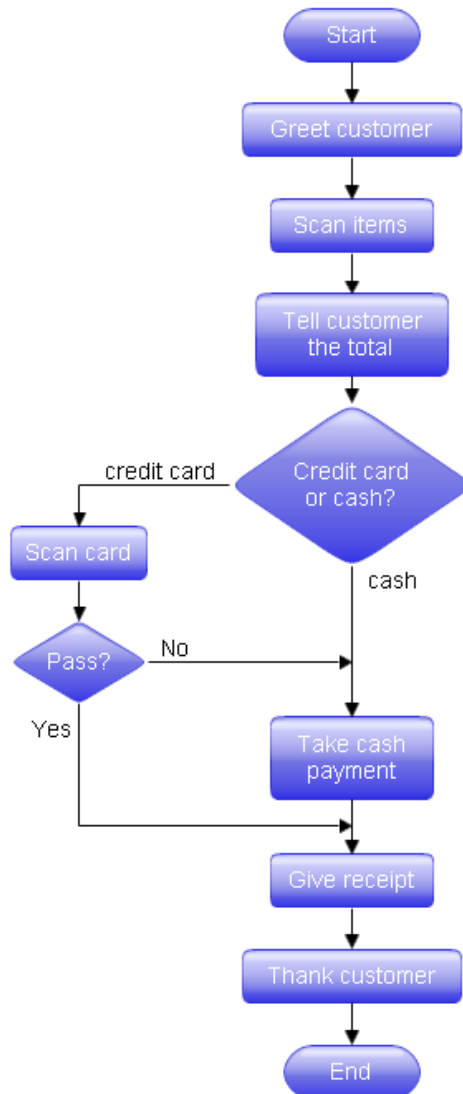
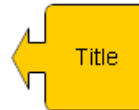
Beverages

C	Input +		Output +	Annotation
	Dish	Guests with children	Beverages	
1	"Spareribs"	-	"Aecht Schlenkerla Rauchbier"	Tough Stuff
2	"Stew"	-	"Guinness"	-
3	"Roastbeef"	-	"Bordeaux"	-
4	"Steak", "Dry Aged Gourmet Steak", "Light Salad and a nice Steak"	-	"Pinot Noir"	-
5	-	true	"Apple Juice"	-
6	-	-	"Water"	-
+	-	-	-	-



Flowchart

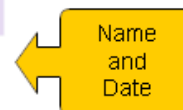
Store Check-out Process



https://www.rff.com/flowchart_shapes.php



J. Smith 8/9/2009



ANSI/ISO Shape	Name
	Flowline (Arrowhead) ^[15]
	Terminal ^[14]
	Process ^[15]
	Decision ^[15]
	Input/Output ^[15]
	Annotation ^[14] (Comment) ^[15]
	Predefined Process ^[14]
	On-page Connector ^[14]
	Off-page Connector ^[14]

DALŠÍ DIAGRAMY

O které můžete zakopnout

Metamodel

- „nad-model“ – model modelu
- Popisuje z jakých prvků se model může skládat a jaké vztahy jsou mezi jednotlivými prvky modelu přípustné.
- Každý slušný modelovací jazyk má metamodel, tak aby výsledný model nebyl jen spontánní obrázek
- Pro zachycení se používá UML diagram tříd

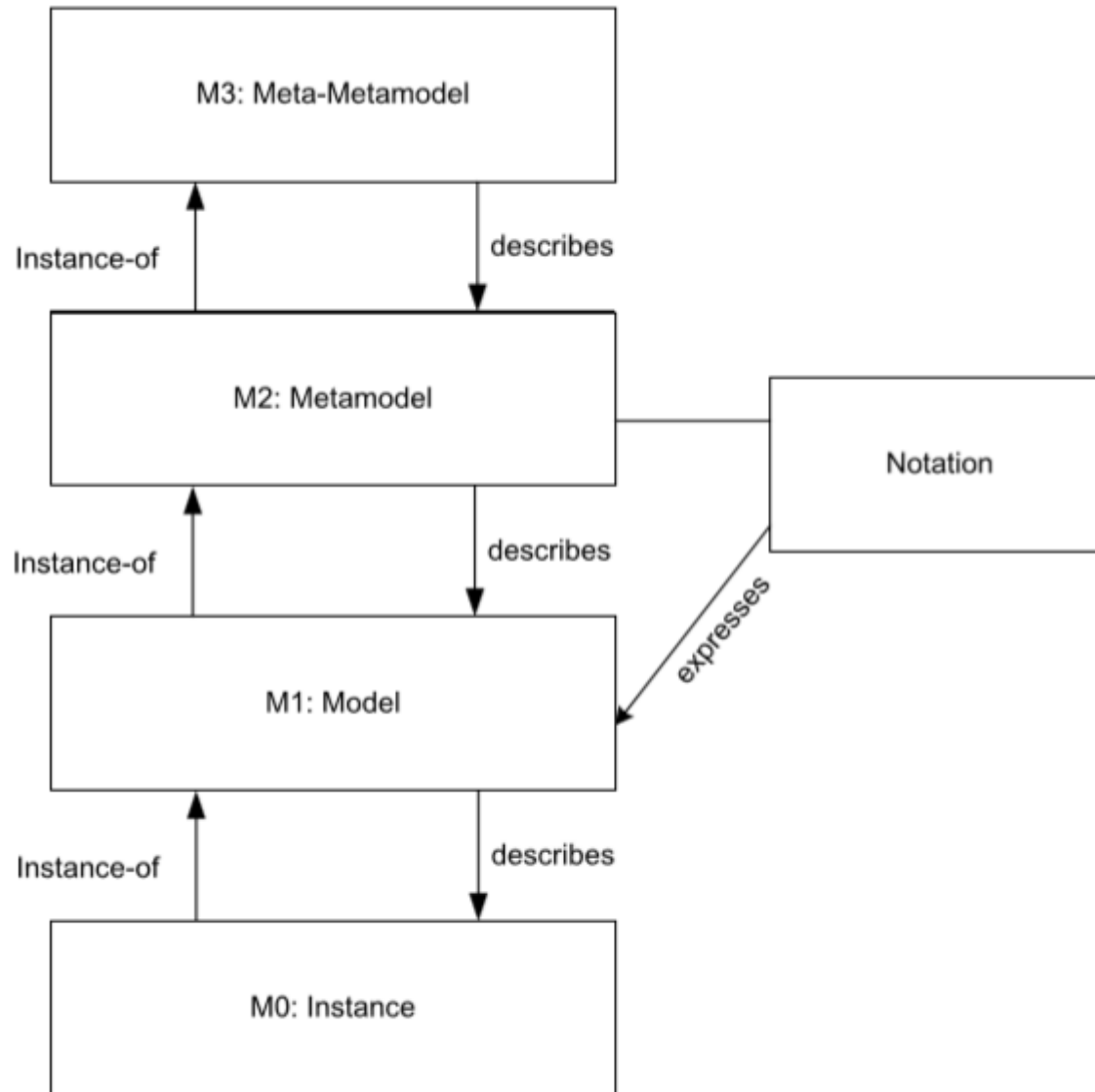
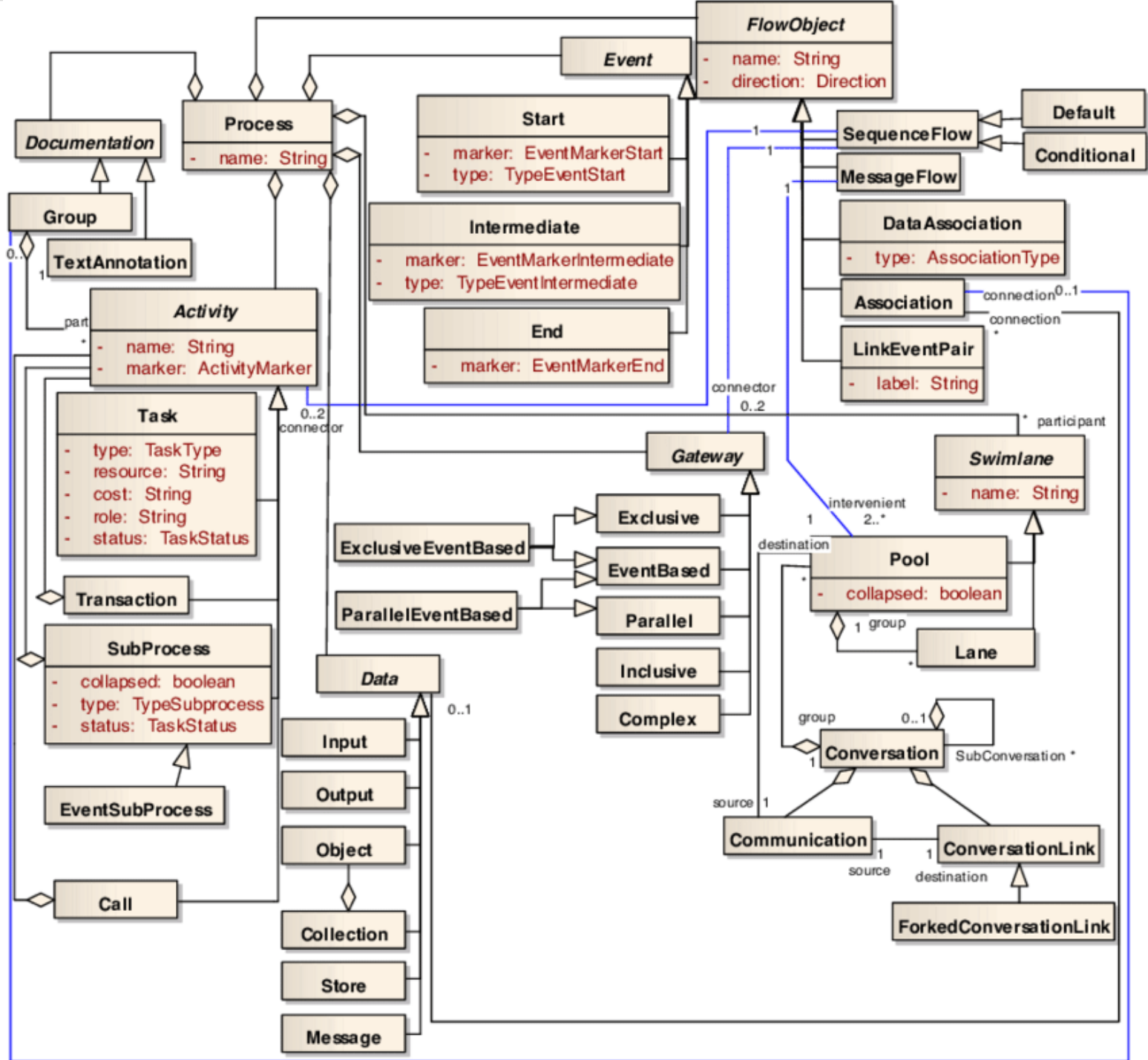
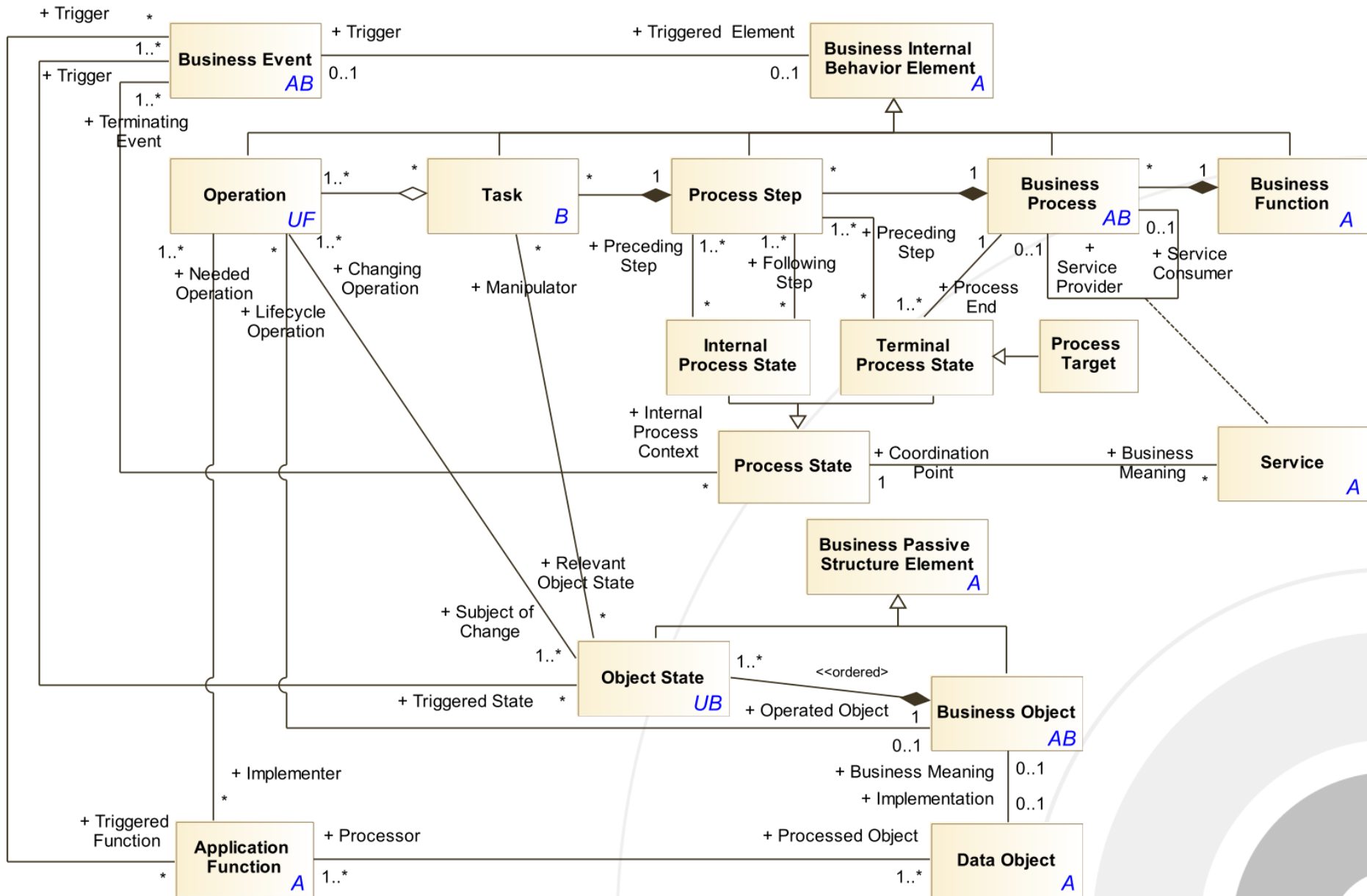


Fig. 3.2. Levels of abstraction

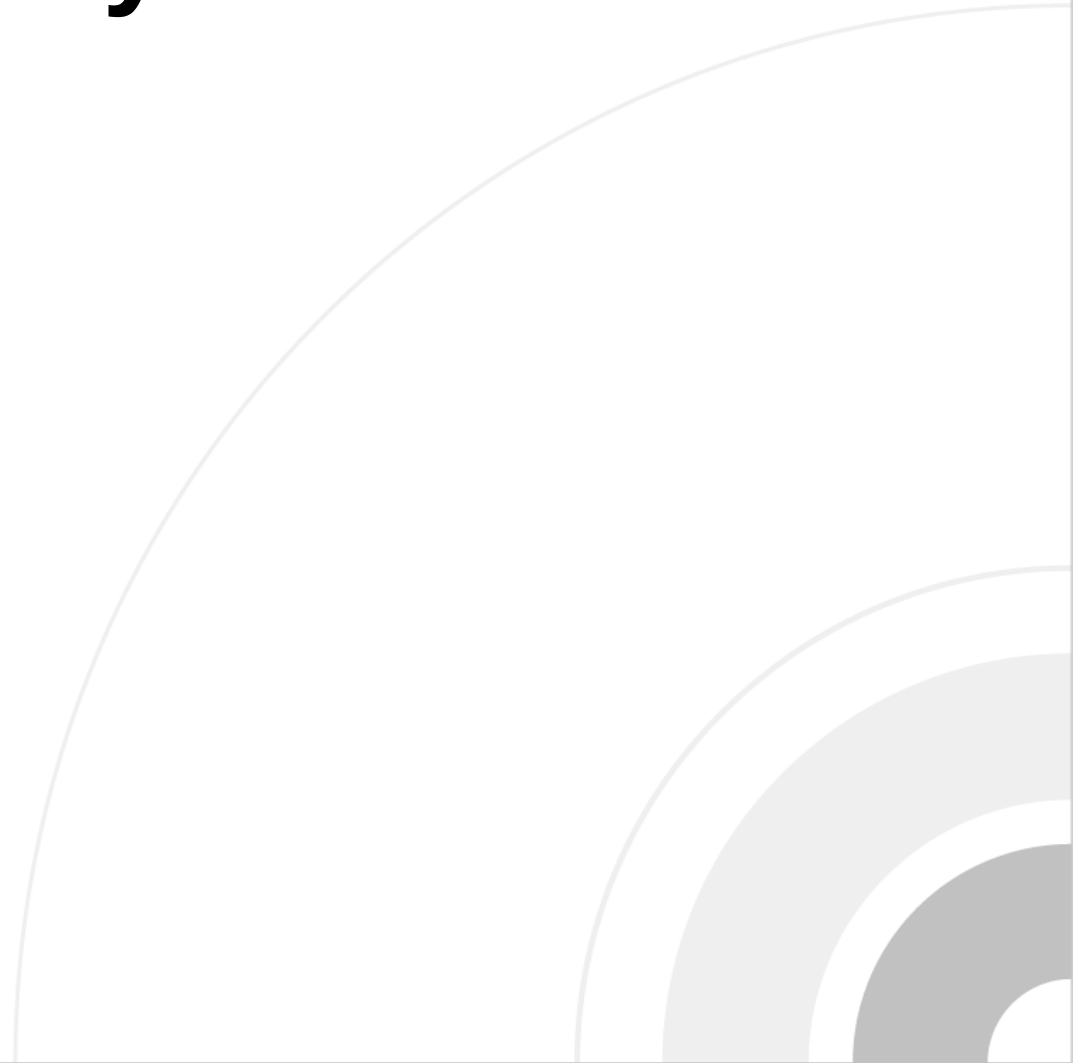
Meta model BPMN

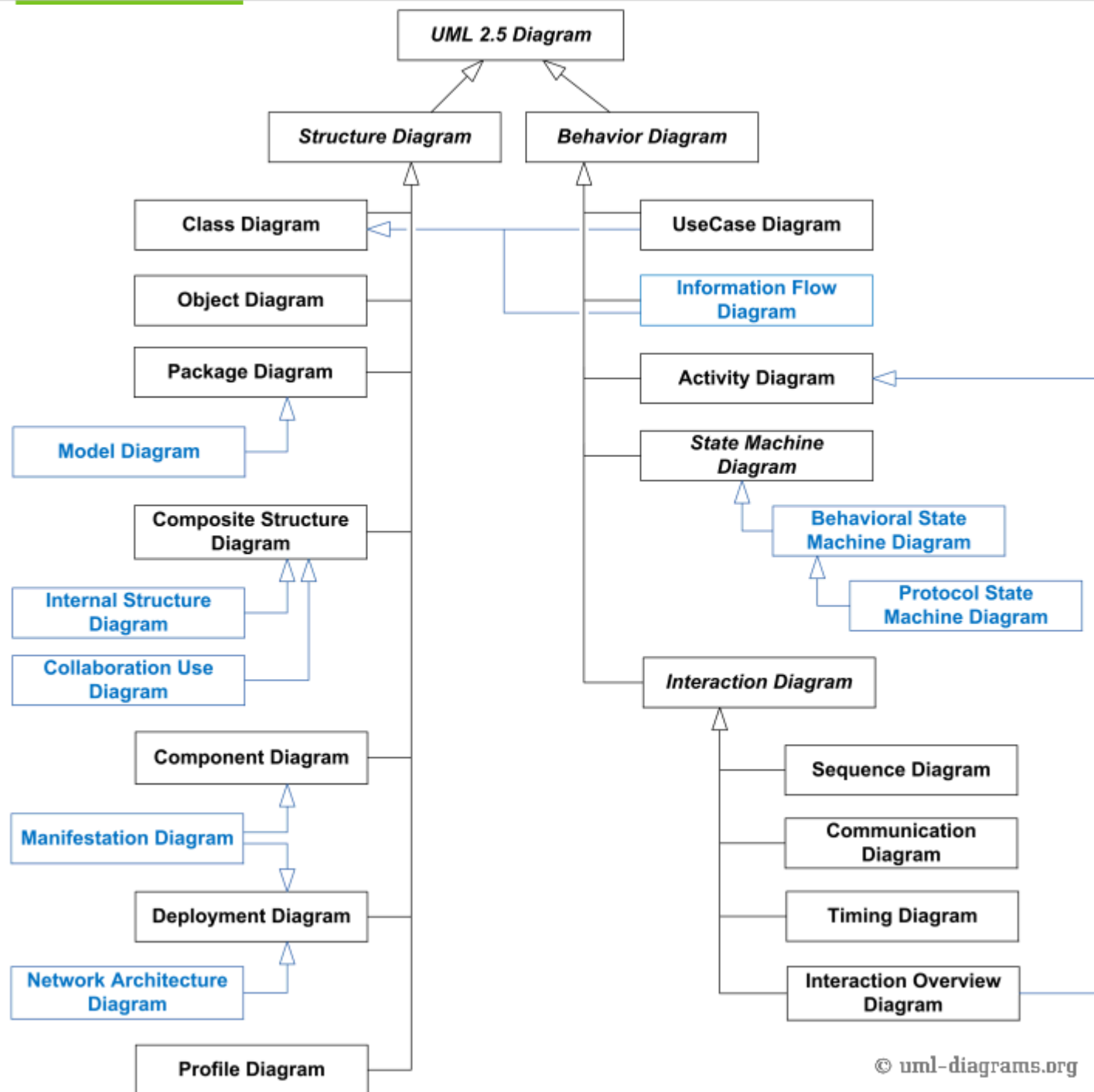


ČÁST MMABP

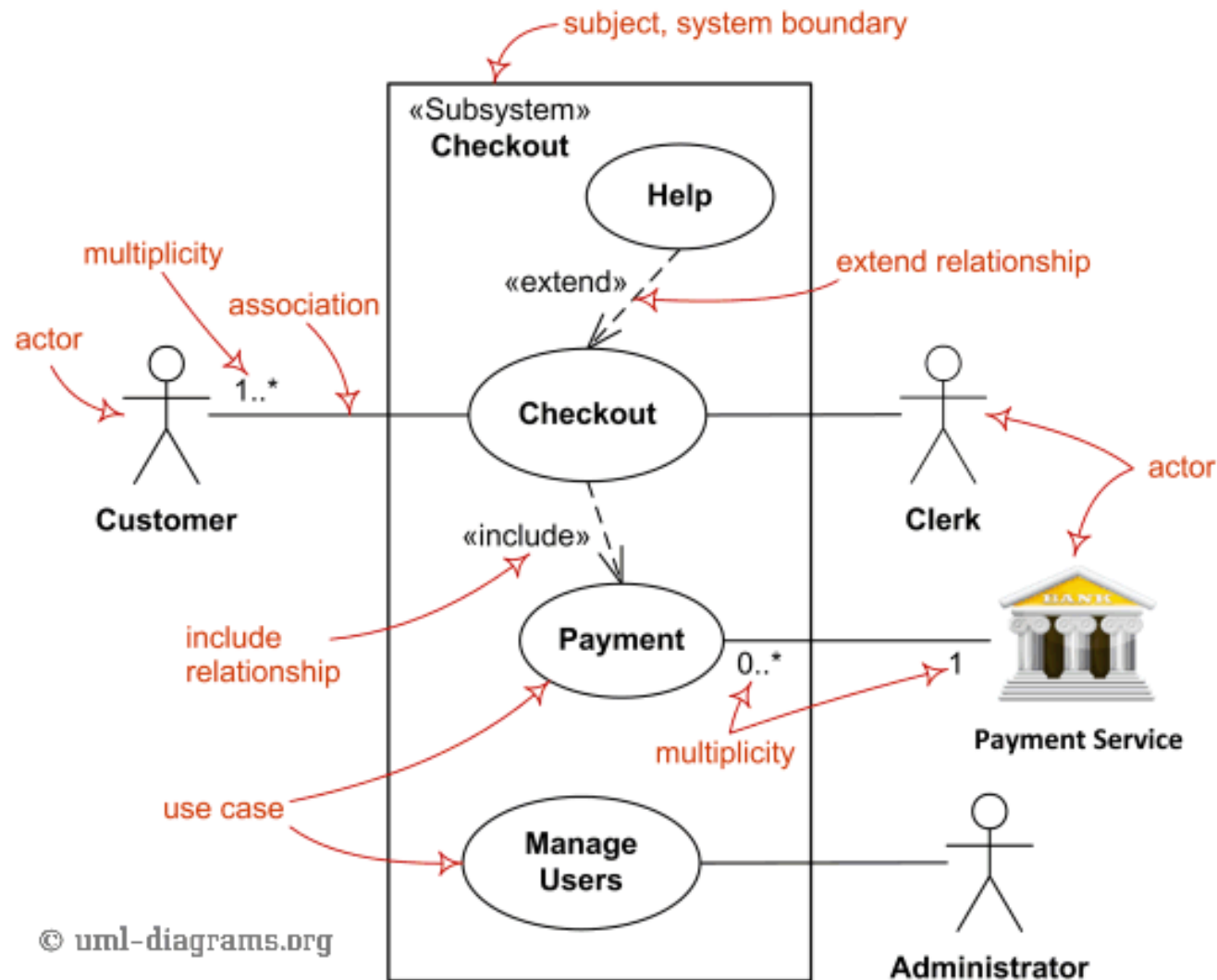


UML Diagramy



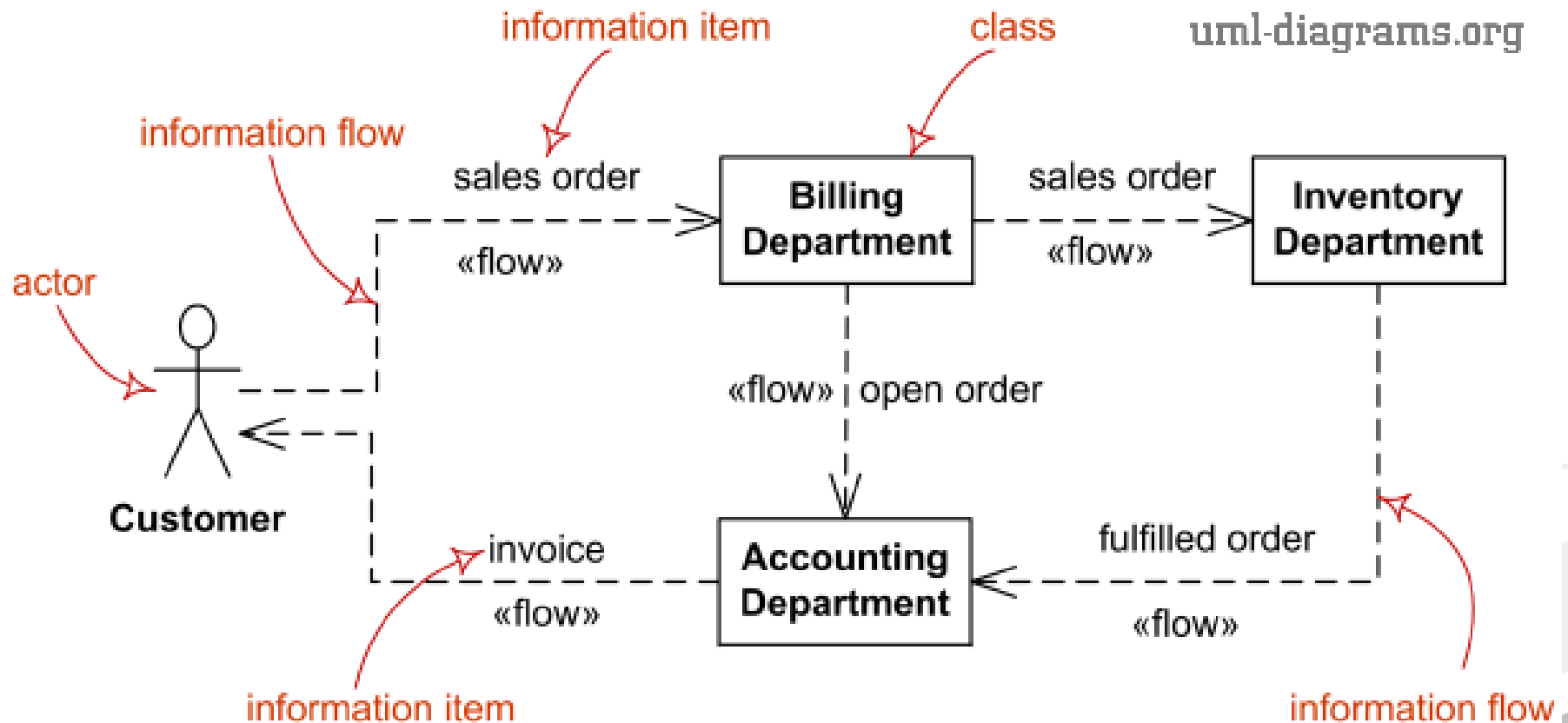


Use case

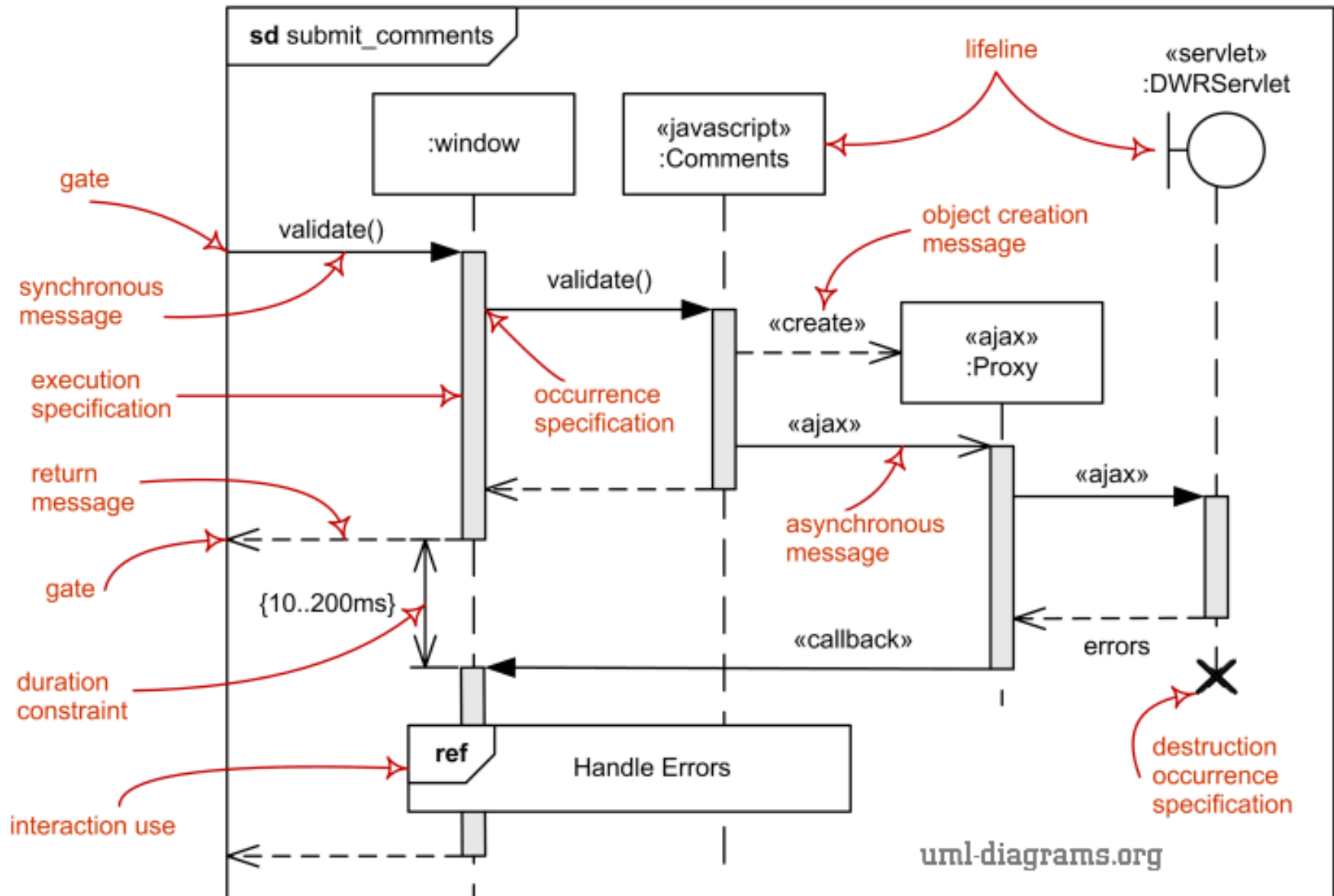


UML 2.5 Information Flow Diagram

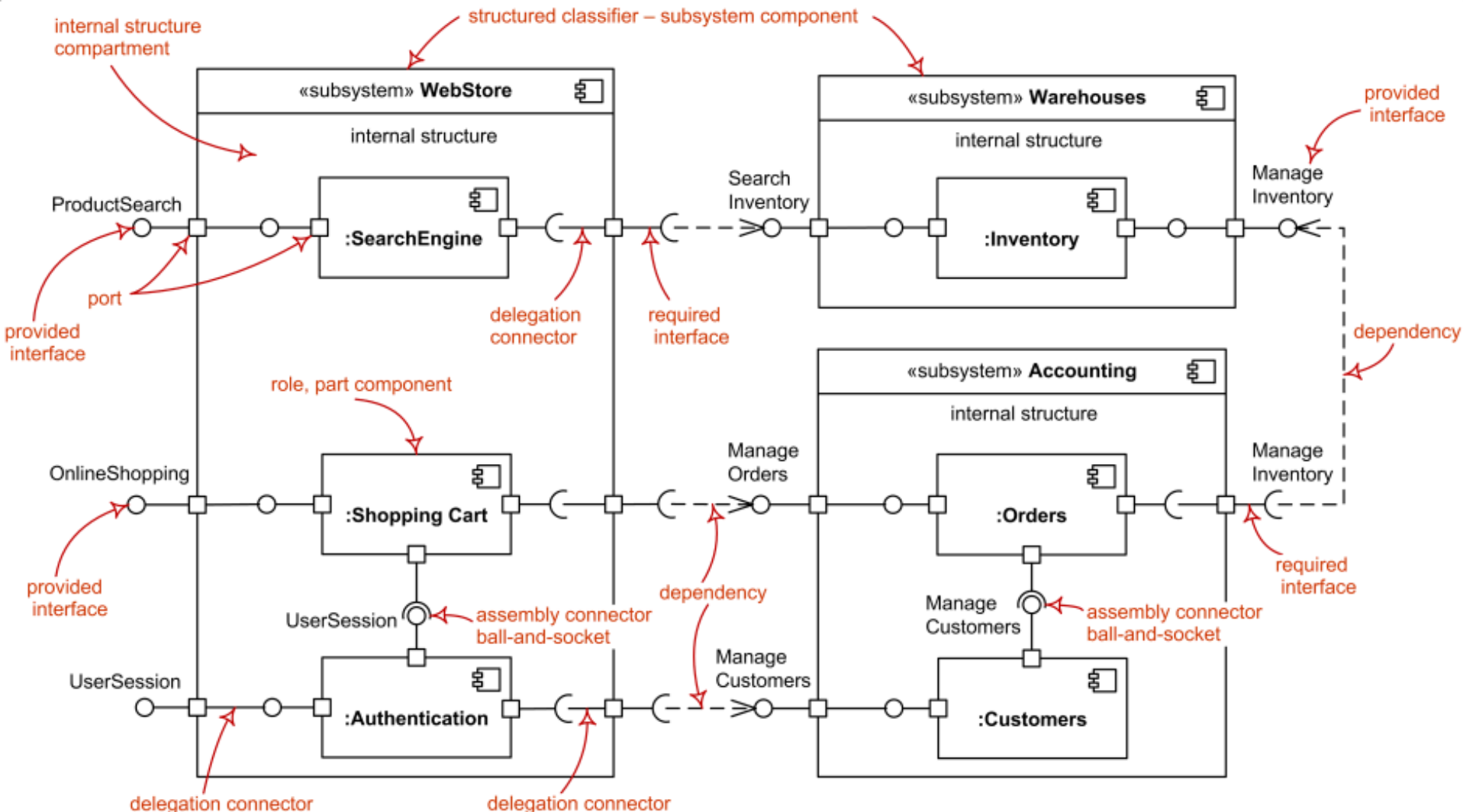
uml-diagrams.org



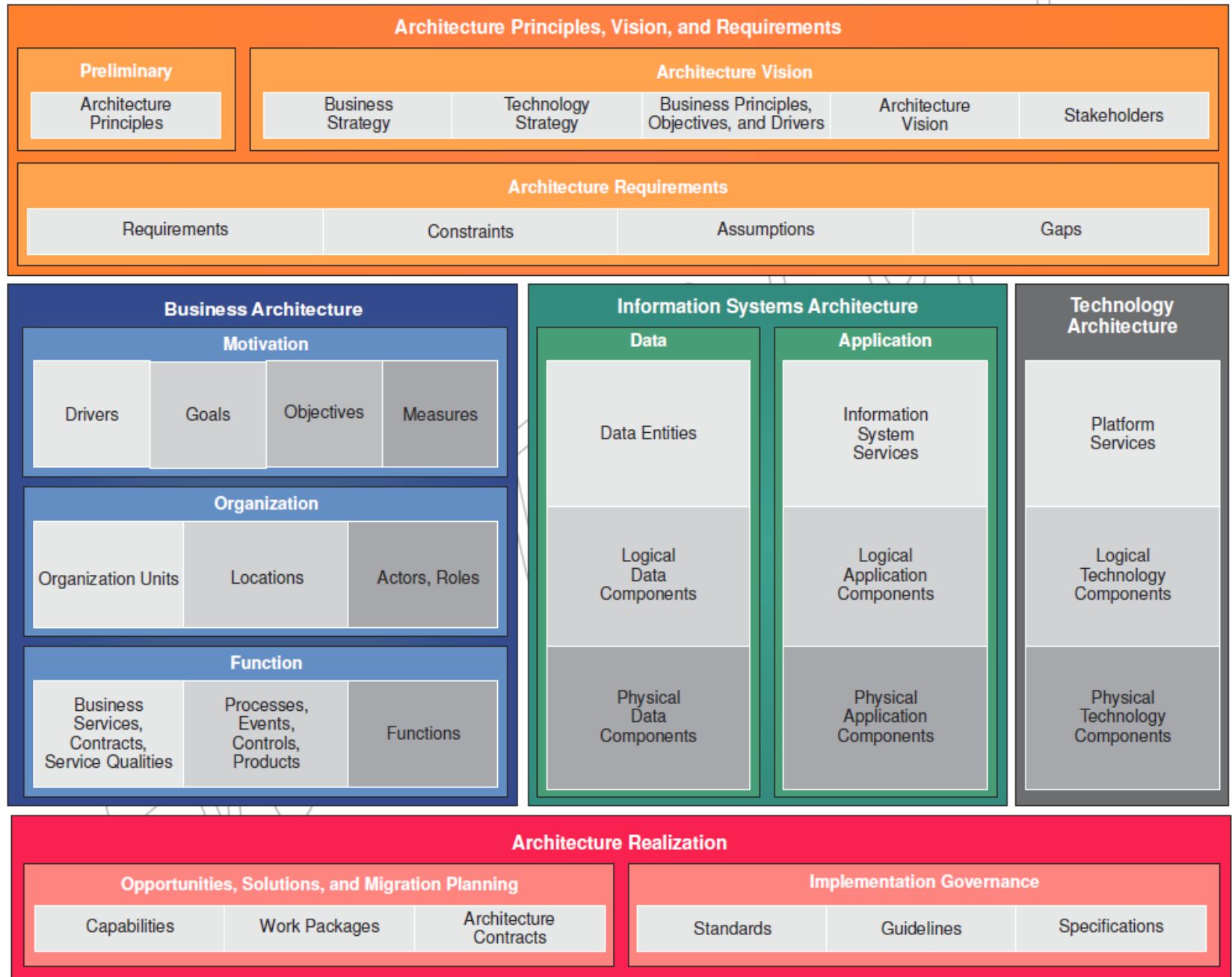
UML Sequence Diagram



UML Component Diagram



TOGAF/ ARCHIMATE



- TOGAF

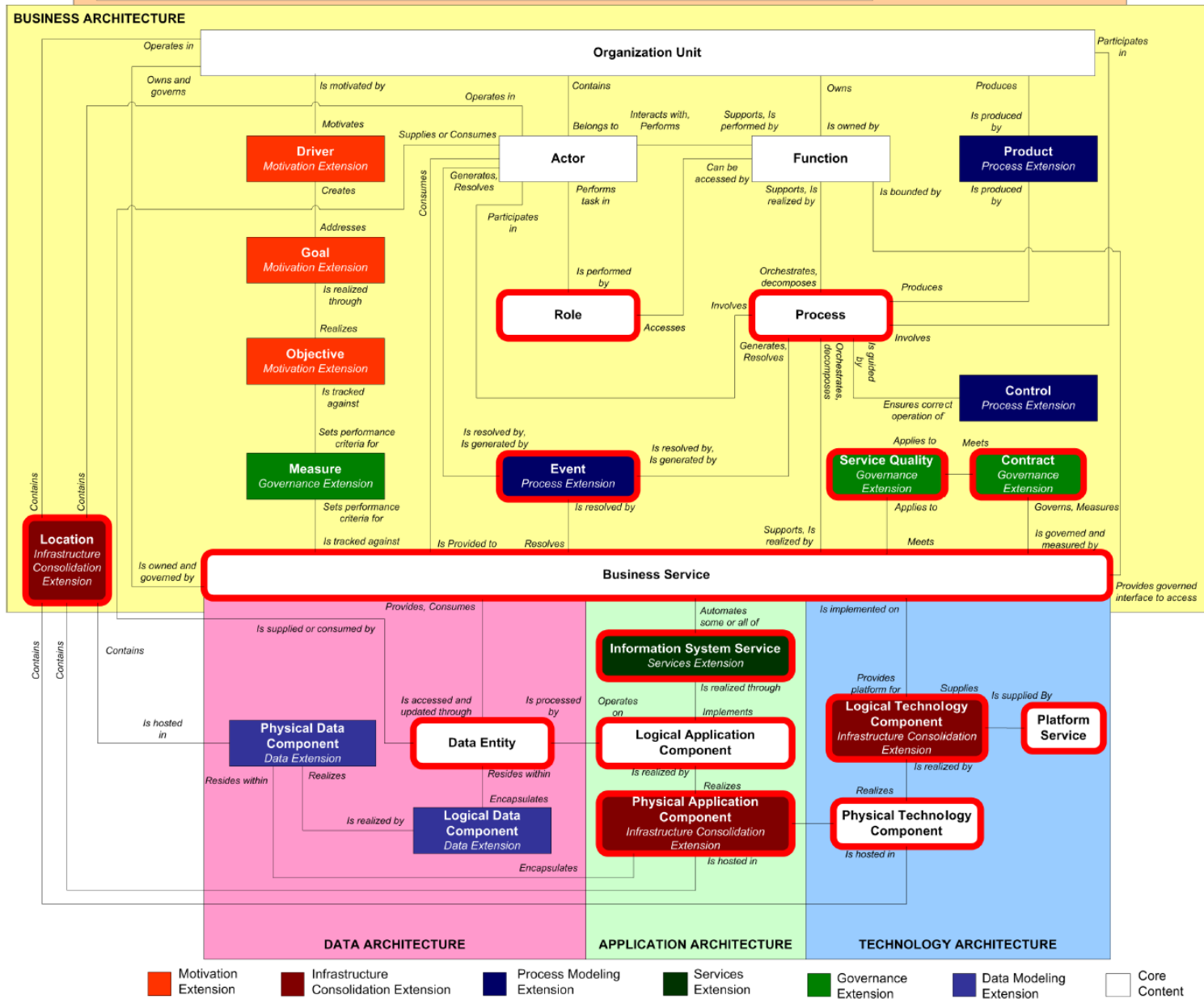
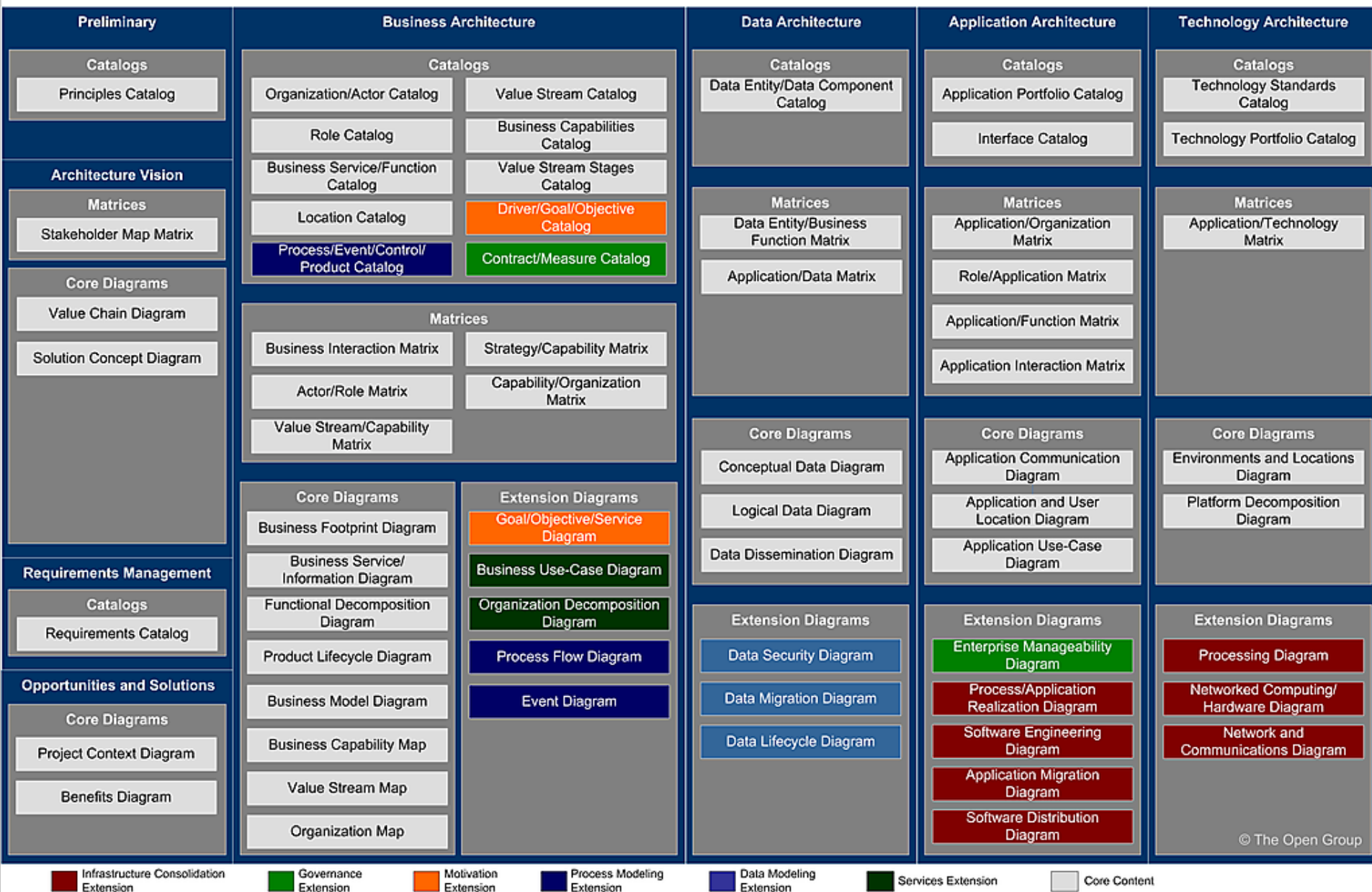


Figure 22-3 SOA Entities in the Content Metamodel

TOGAF Diagrams



Infrastructure Consolidation Extension

Governance Extension

Motivation Extension

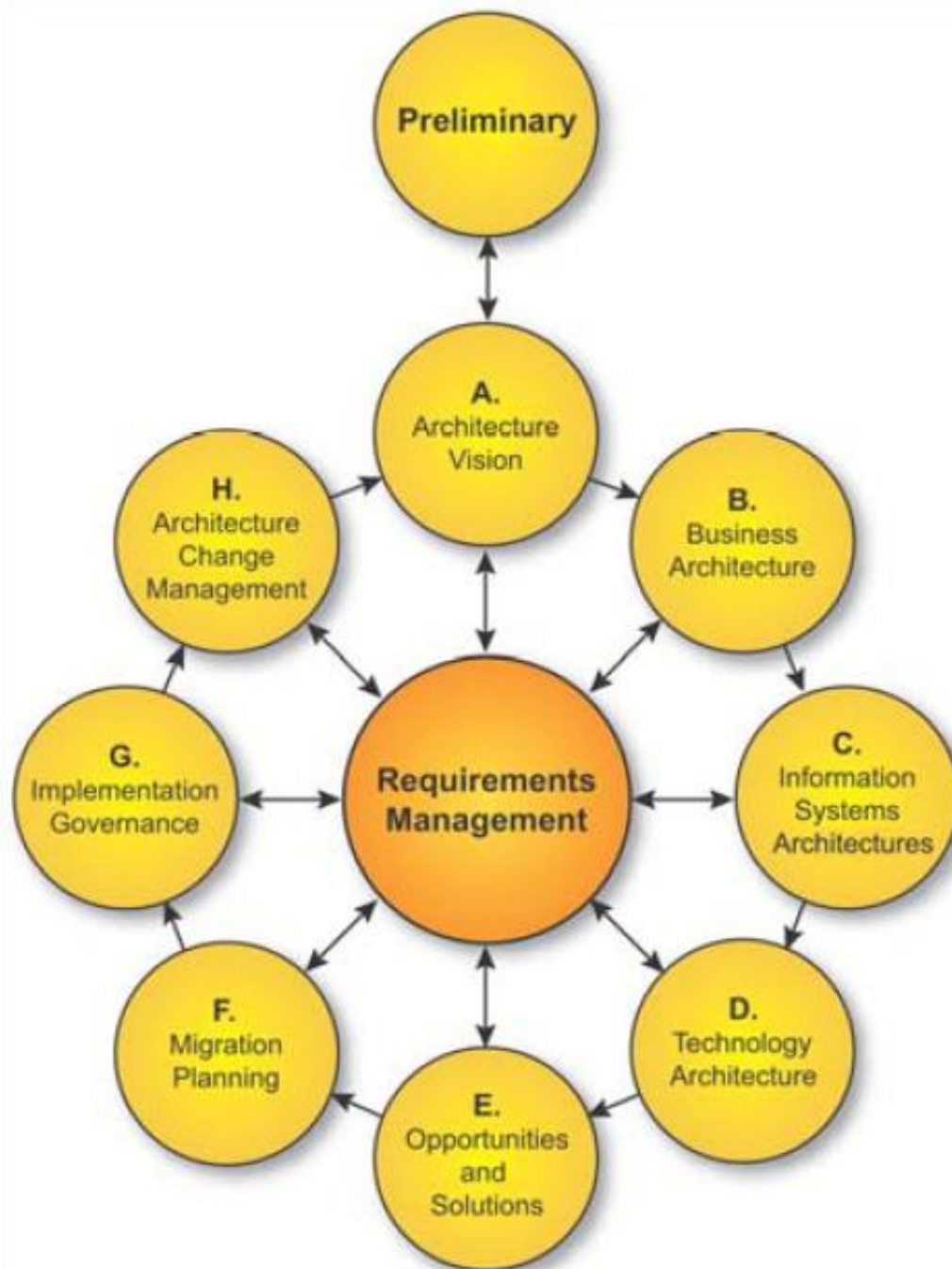
Process Modeling Extension

Data Modeling Extension

Services Extension

Core Content

Introduction to the Architecture Development Method (ADM)



TOGAF is a registered trademark of The Open Group in the United States and other countries

TOGAF®

What is the TOGAF ADM?

- The ADM forms the core of TOGAF
- The result of contributions from many architecture practitioners
- A process for developing an enterprise architecture
- Integrates all the elements within TOGAF
- Designed to address enterprise's business and IT needs by providing:
 - A set of architecture views (business, data, application, technology)
 - A set of recommended deliverables
 - A method for managing requirements
 - Guidelines on tools for architecture development

Prepare the organization for a successful architecture project

Provide continual monitoring and a change management process to ensure that the architecture responds to the needs of the enterprise

Provide architectural oversight for the implementation; ensure that the implementation project conforms to the architecture

Analyze costs, benefits and risks; develop detailed Implementation and Migration Plan

Ensure that every stage of a TOGAF project is based on and validates business requirements

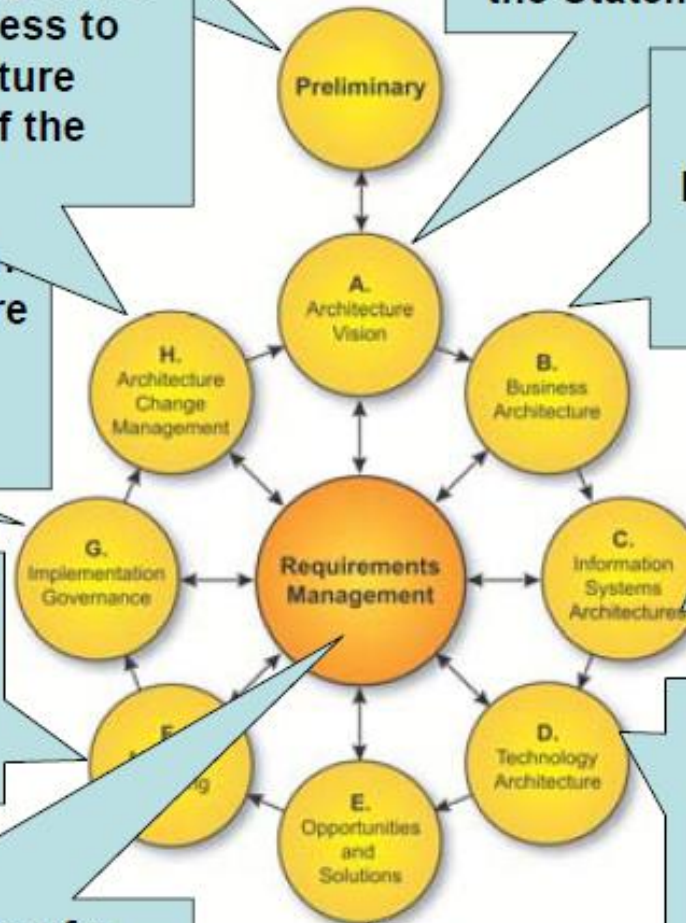
Set the scope, constraints and expectations for a TOGAF project; create the Architecture Vision; validate the business context; create the Statement of Architecture Work

Develop Business Architecture
Develop baseline and target architectures and analyze the gaps

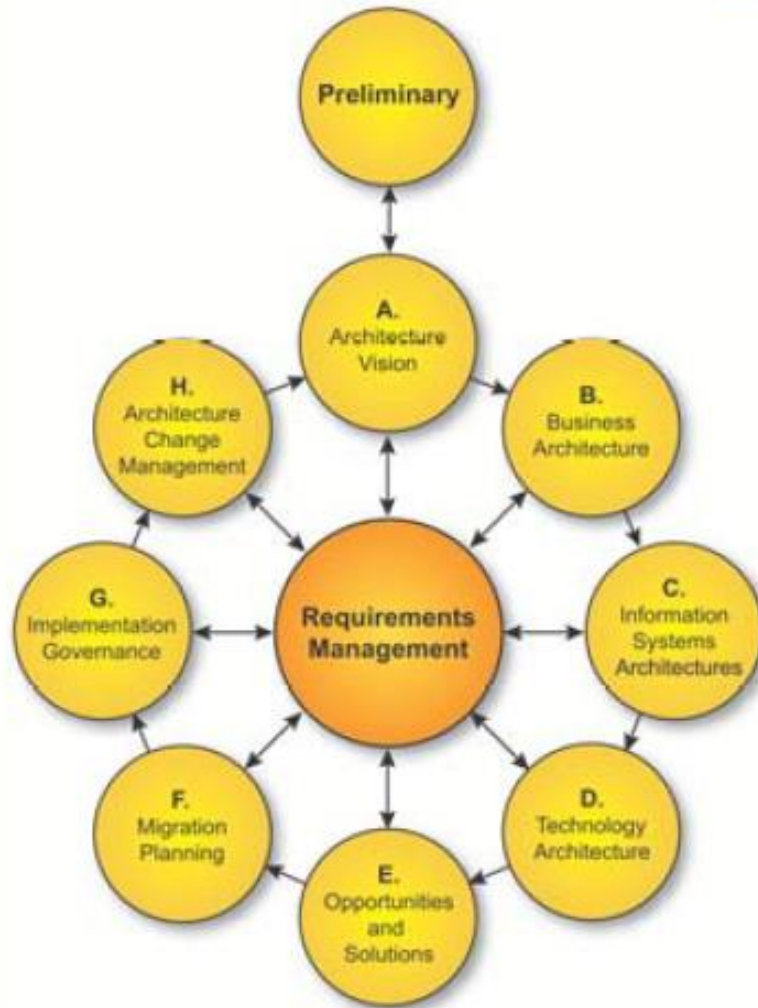
Develop Information Systems Architectures
Develop baseline and target architectures and analyze the gaps

Develop Technology Architecture
Develop baseline and target architectures and analyze the gaps

Perform initial implementation planning; identify major implementation projects



Architecture Development Method – Process



- The ADM is an iterative process:
 - Over the whole process
 - Between phases
 - Within phases
- For each iteration, re-consider:
 - Scope
 - Detail
 - Schedules, milestones

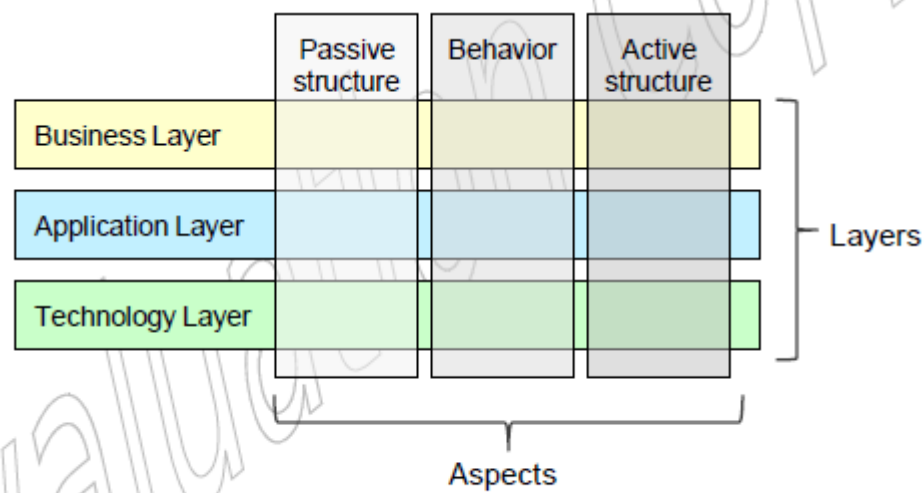


Figure 2: ArchiMate Core Framework

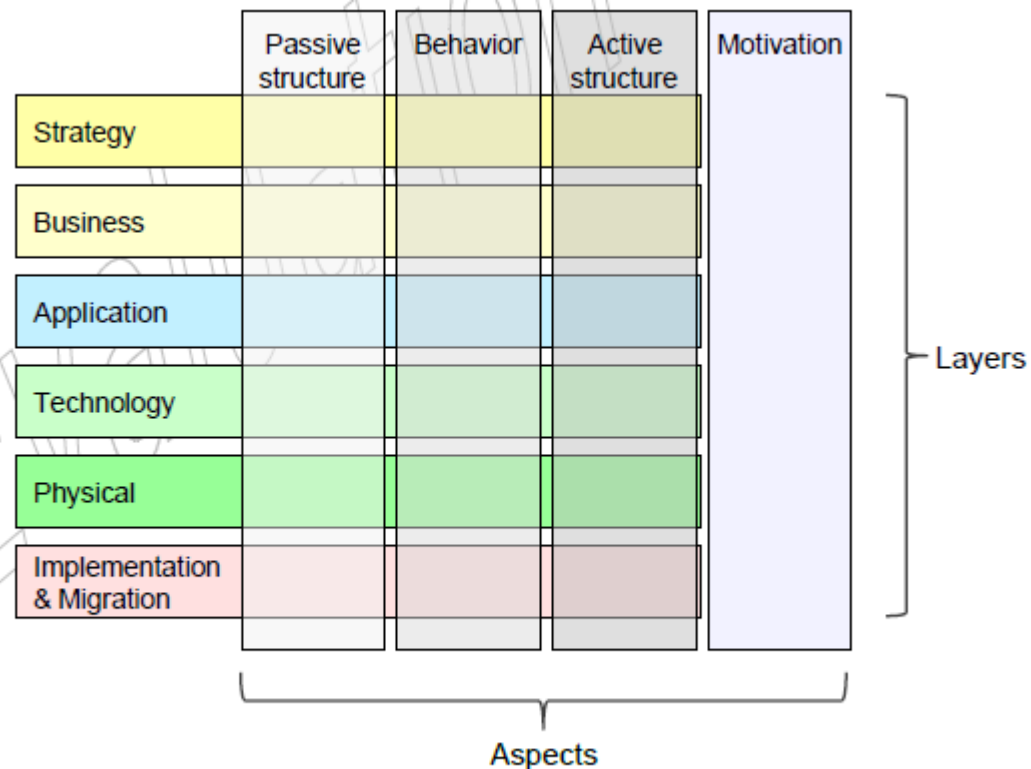


Figure 3: Full ArchiMate Framework

The Open Group. (2016). ArchiMate 3.0 Specification. Van Haren. ISBN 94-01-80047-2

MetaModel

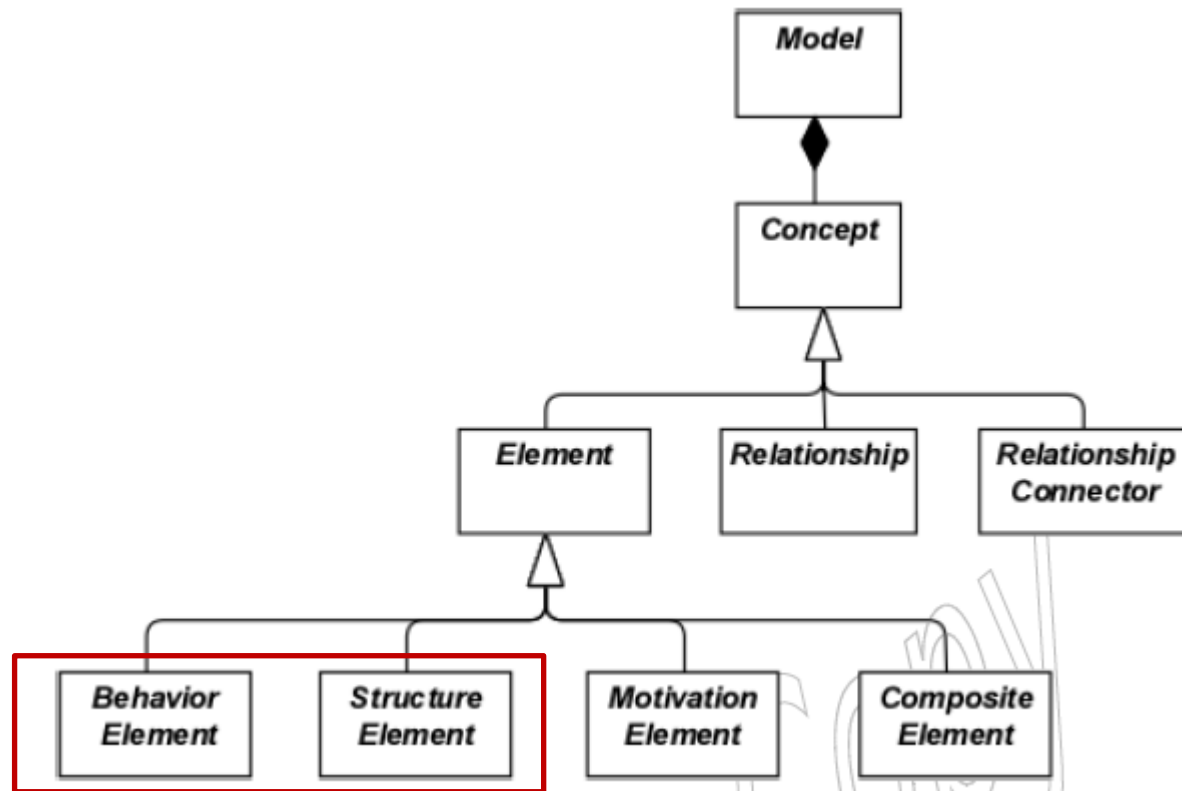


Figure 1: Top-Level Hierarchy of ArchiMate Concepts

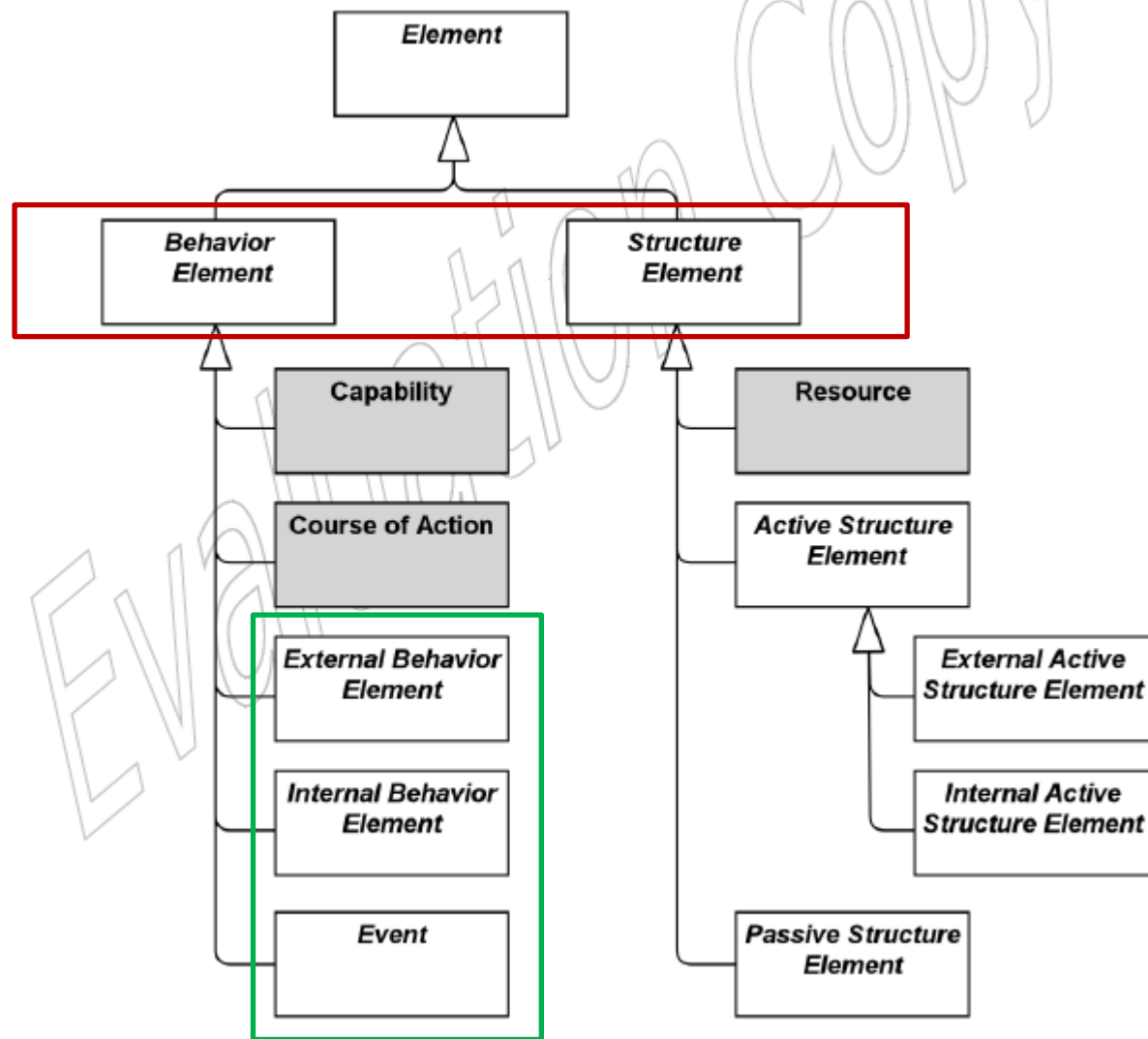


Figure 4: Hierarchy of Behavior and Structure Elements

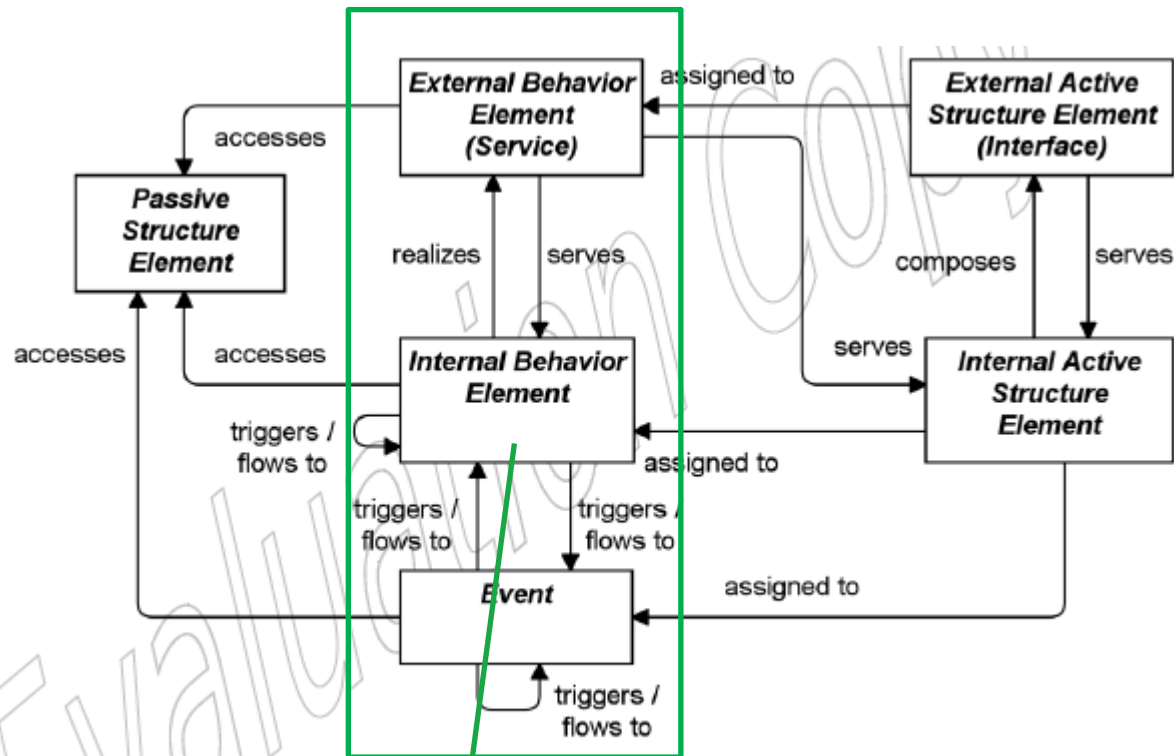


Figure 5: Behavior and Structure Elements Metamodel

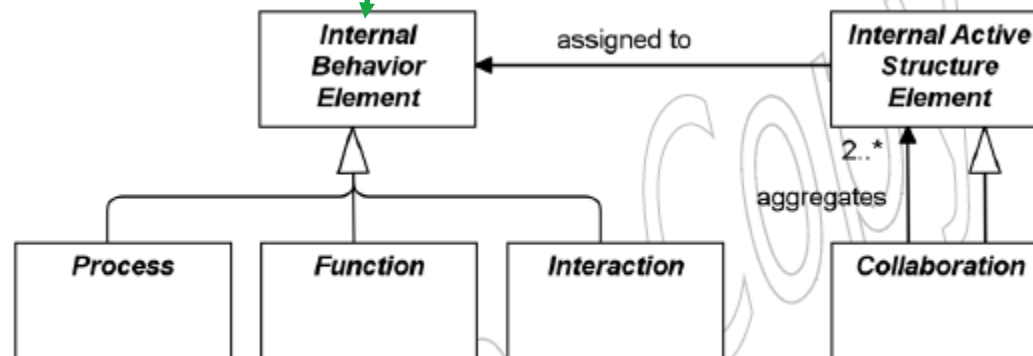
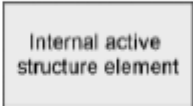
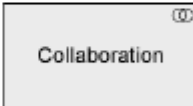
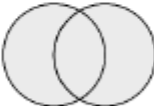
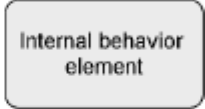
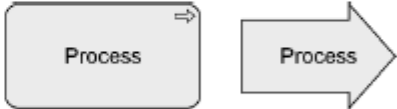
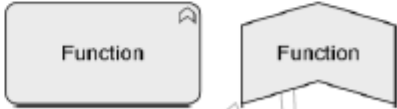
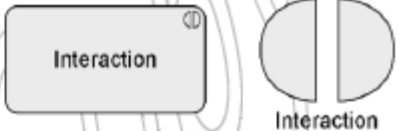
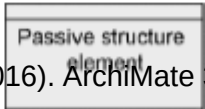


Figure 12: Specializations of Core Elements

Notace

Table 1: Core Elements

Element	Specializations	Definition	Notation
Active Structure			
Internal active structure element		An entity that is capable of performing behavior.	
	Collaboration	An aggregate of two or more active structure elements, working together to perform some collective behavior.	  Collaboration
Interface		A point of access where one or more services are exposed available to the environment.	  Interface

Element	Specializations	Definition	Notation
Behavior			
Internal behavior element		A unit of activity performed by one or more active structure elements.	
	Process	A sequence of behaviors that achieves a specific outcome.	
	Function	A collection of behavior based on specific criteria, such as required resources, competences, or location.	
	Interaction	A unit of collective behavior performed by (a collaboration of) two or more structure elements.	
Service		An explicitly defined exposed behavior.	
Event		A state change.	
Passive Structure			
Passive structure element		An element on which behavior is performed.	

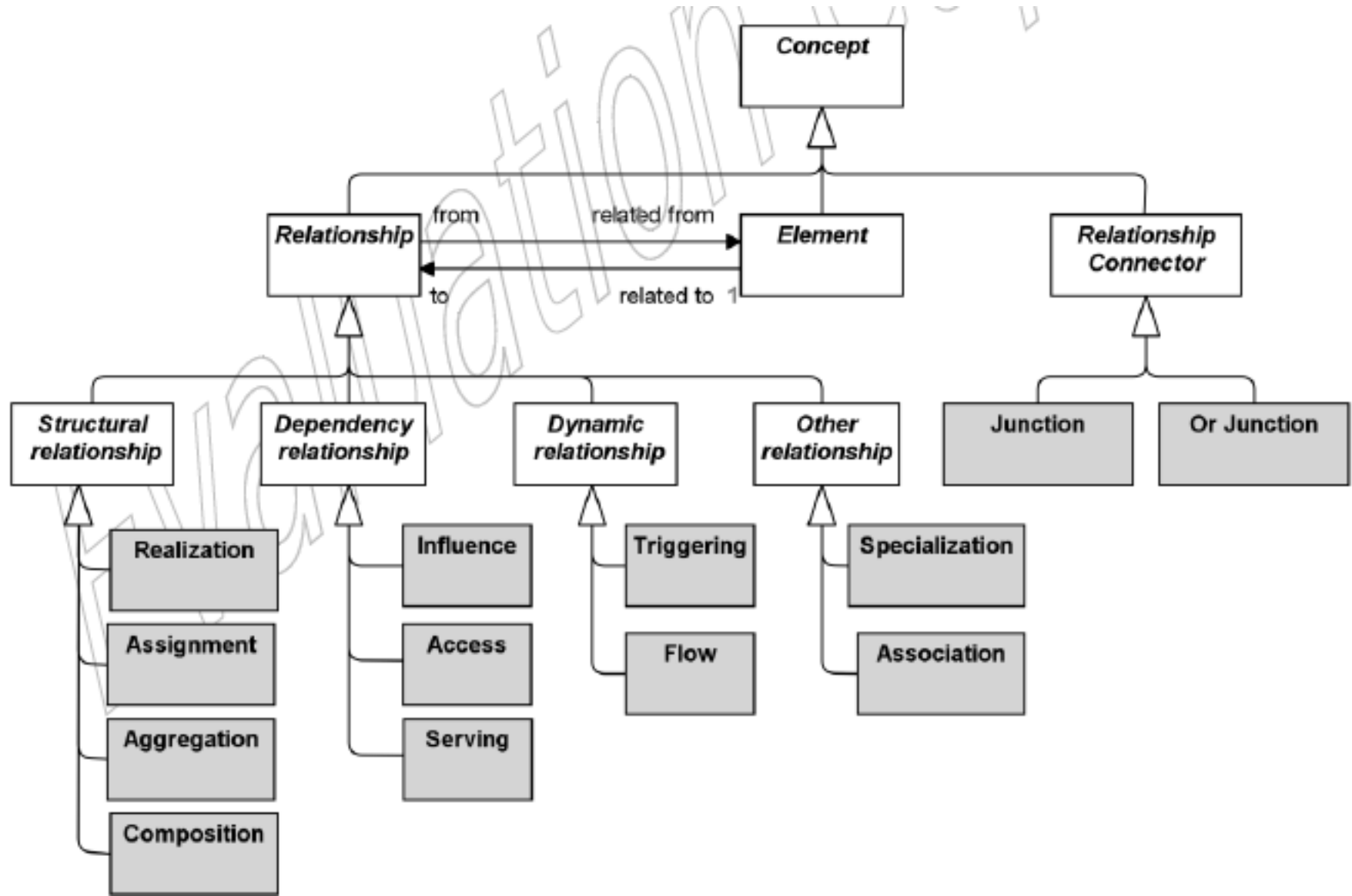
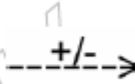


Figure 19: Overview of Relationships

Notace

Table 3: Relationships

Structural Relationships		Notation
Composition	Indicates that an element consists of one or more other elements.	
Aggregation	Indicates that an element groups a number of other elements.	
Assignment	Expresses the allocation of responsibility, performance of behavior, or execution.	
Realization	Indicates that an entity plays a critical role in the creation, achievement, sustenance, or operation of a more abstract entity.	

Dependency Relationships		Notation
Serving	Models that an element provides its functionality to another element.	
Access	Models the ability of behavior and active structure elements to observe or act upon passive structure elements.	  
Influence	Models that an element affects the implementation or achievement of some motivation element.	
Dynamic Relationships		Notation
Triggering	Describes a temporal or causal relationship between elements.	
Flow	Transfer from one element to another.	
Other Relationships		Notation
Specialization	Indicates that an element is a particular kind of another element.	
Association	Models an unspecified relationship, or one that is not represented by another ArchiMate relationship.	
Junction	Used to connect relationships of the same type.	  (And) Junction Or Junction

Example of Business Layer

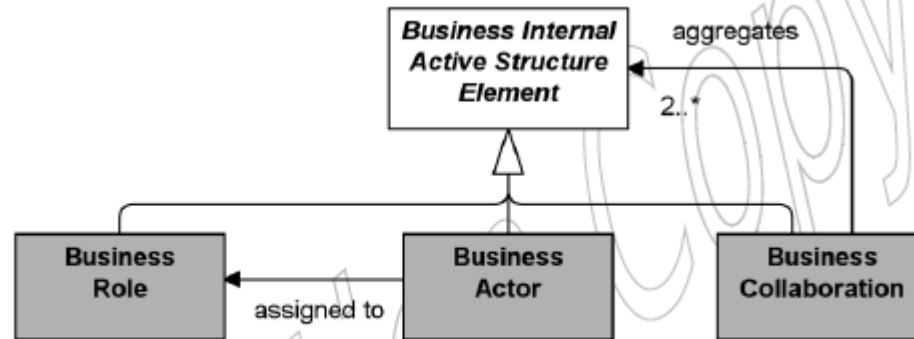


Figure 50: Business Internal Active Structure Elements

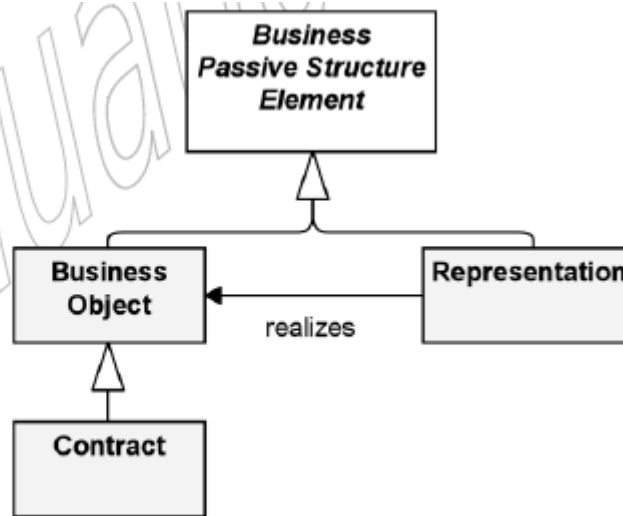


Figure 61: Business Passive Structure Elements

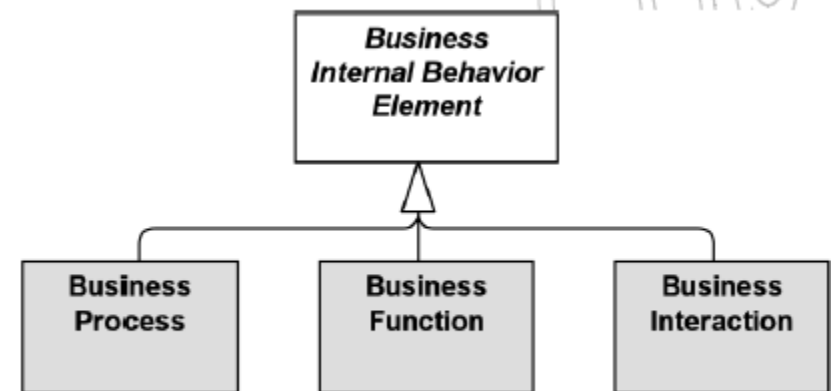
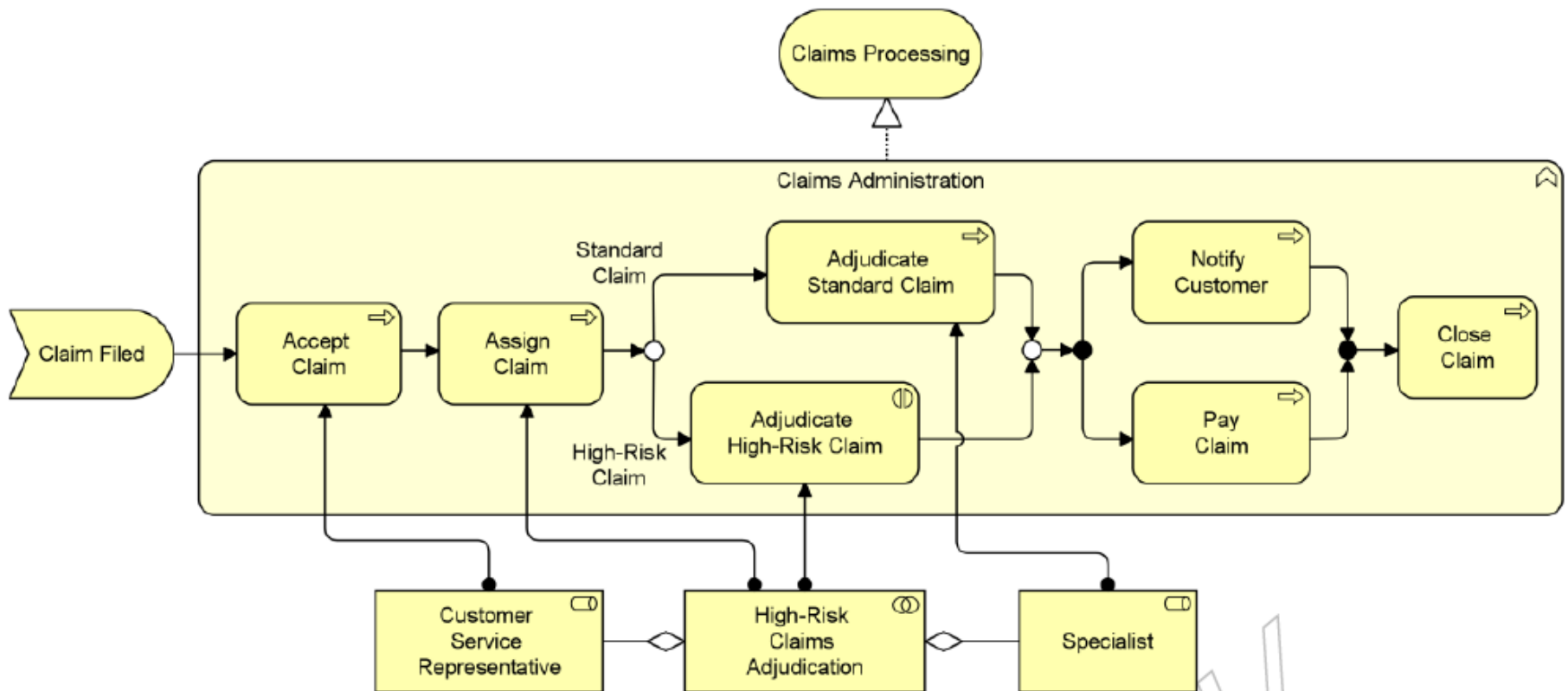


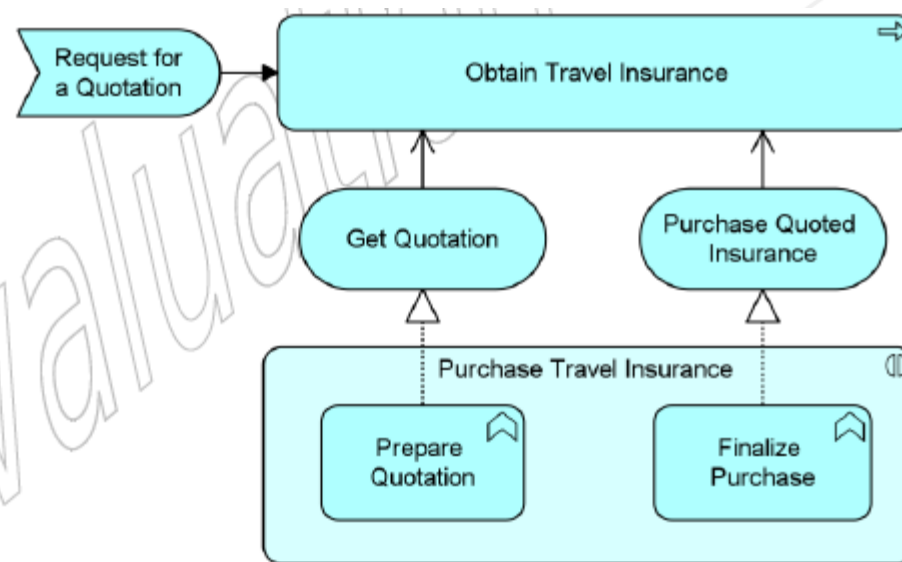
Figure 55: Business Internal Behavior Elements

Business Layer



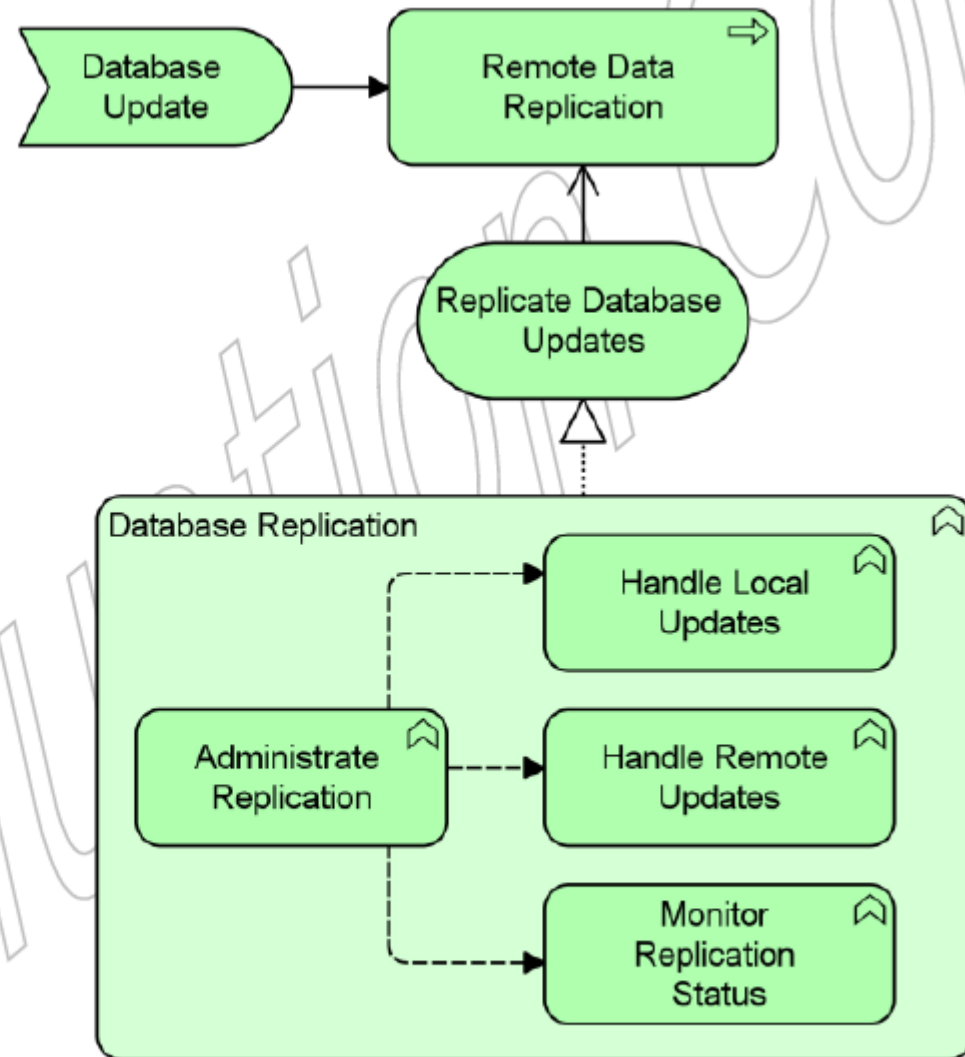
Example 23: Business Behavior Elements

Application Layer



Example 27: Application Behavior Elements

Technology Layer



Example 30: Technology Behavior Elements

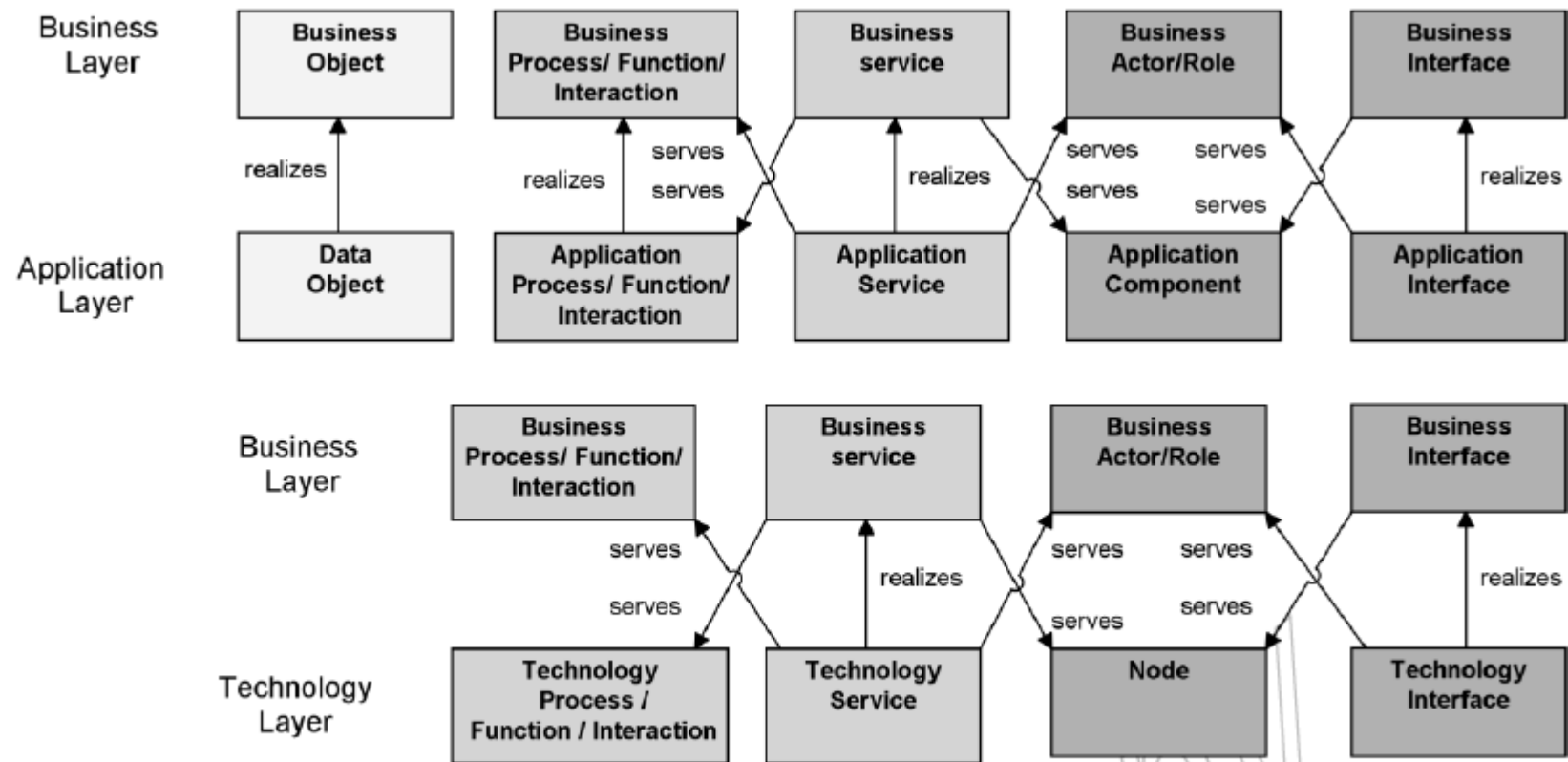


Figure 96: Relationships between Business Layer and Application and Technology Layer Elements

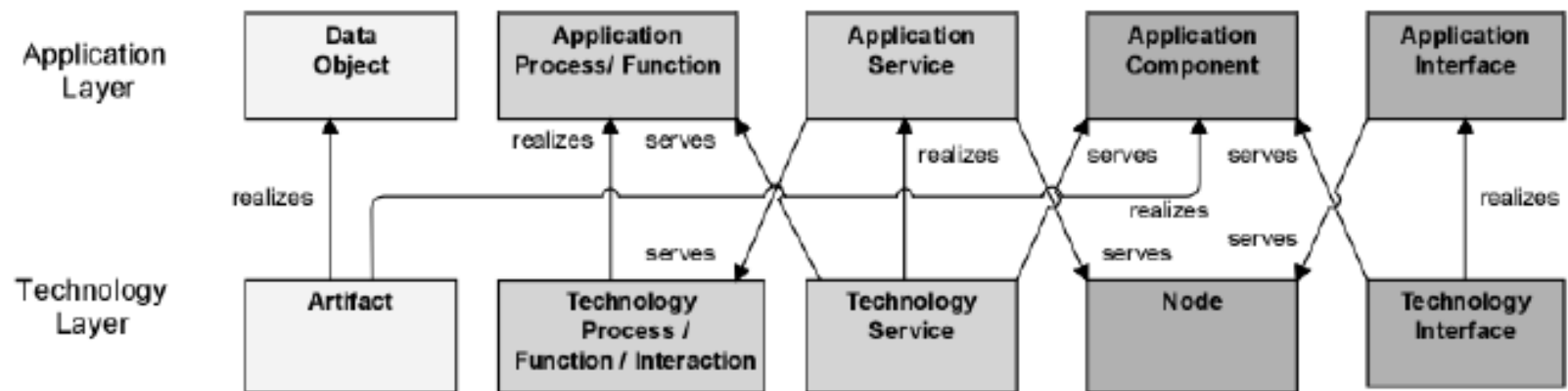
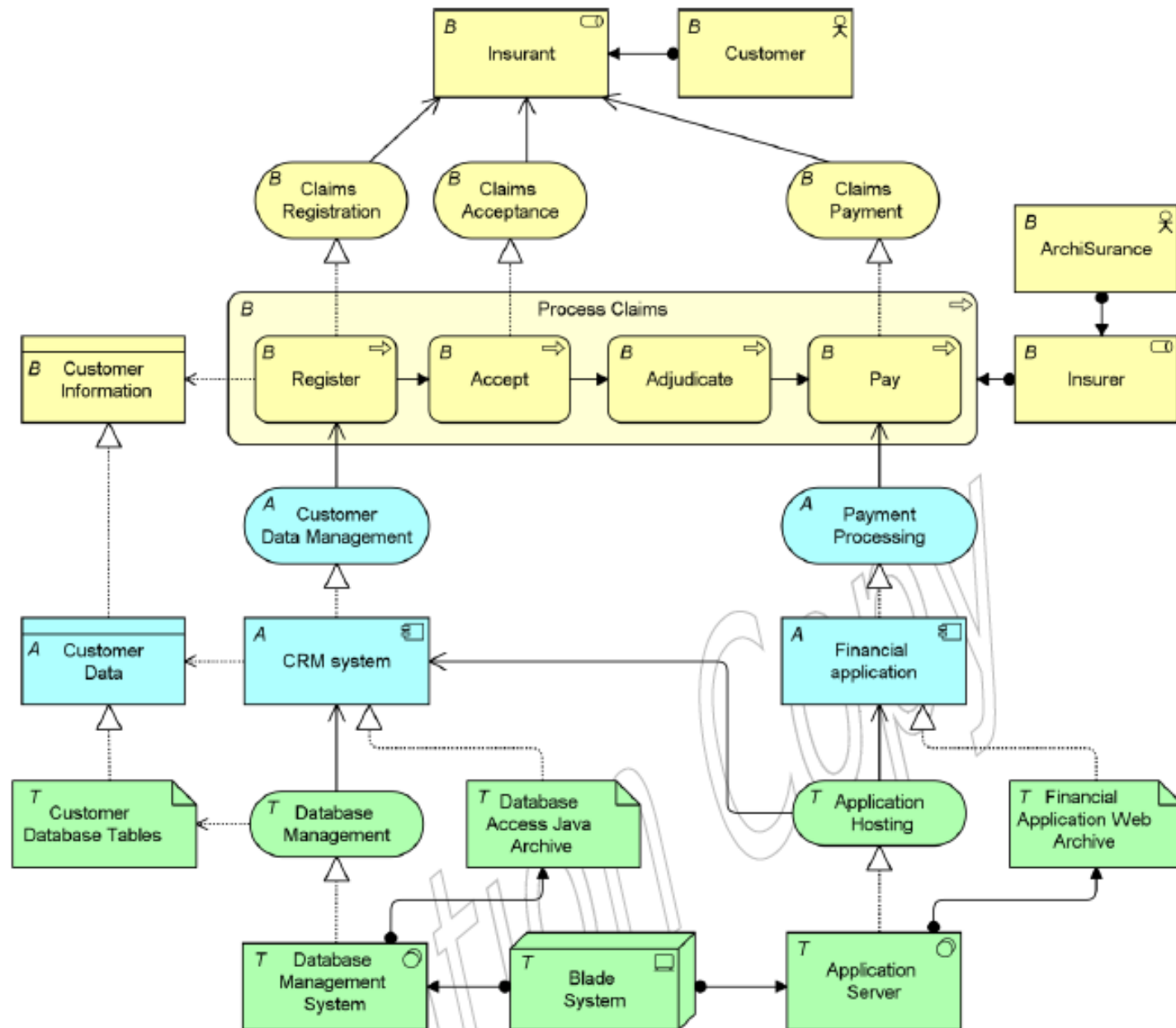


Figure 97: Relationships between Application Layer and Technology Layer Elements



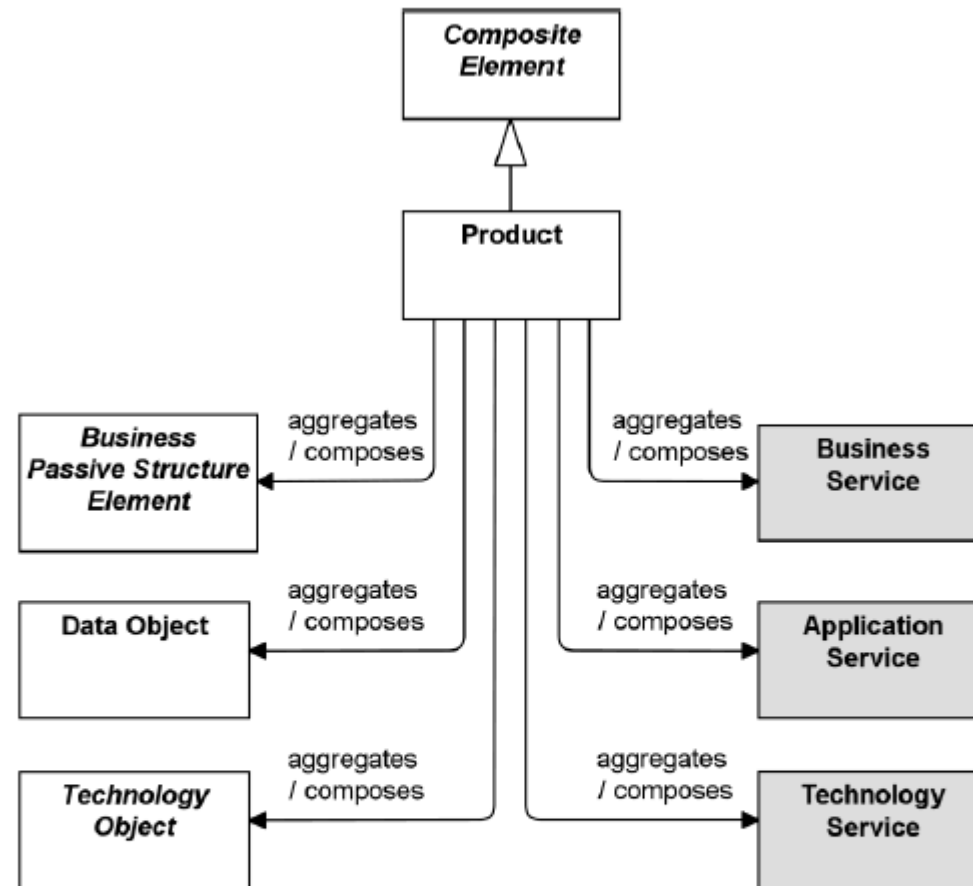


Figure 65: Product

Product

A product represents a coherent collection of services and/or passive structure elements, accompanied by a contract/set of agreements, which is offered as a whole to (internal or external) customers.